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A hybrid CNN-DNN model for battery remaining useful life RUL prediction

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Accurate prediction of the Remaining Useful Life of lithium-ion batteries is essential for enhancing reliability, safety, and maintenance planning in energy storage systems. This study proposes an optimized hybrid deep learning framework that integrates Convolutional Neural Networks with Deep Neural Networks for effective battery remaining useful life prediction. The model leverages convolutional neural network layers to automatically extract discriminative features from multivariate battery degradation data, while deep neural network layers model complex nonlinear relationships for precise regression estimation. To further enhance predictive performance and reduce feature redundancy, Binary Particle Swarm Optimization is employed for optimal feature selection. Experiments were conducted using a publicly available lithium-ion battery degradation dataset comprising 680 samples with electrical, thermal, and health-related parameters, including State of Health and remaining useful life indicators. The dataset was divided into training, validation, and testing subsets with proportions of 70%, 15%, and 15%, respectively. A comprehensive performance evaluation was performed using Mean Squared Error, Mean Absolute Percentage Error, Median Absolute Error, Mean Absolute Error, and the coefficient of determination. The proposed hybrid model achieved superior performance with a Mean Squared Error of 0.0141, Mean Absolute Error of 0.0931, Mean Absolute Percentage Error of 0.0142, Median Absolute Error of 0.0739, and a coefficient of determination of 99.01%, significantly outperforming comparative deep learning models. These results demonstrate that the proposed framework provides a robust and accurate solution for lithium-ion battery remaining useful life prediction and supports its potential deployment in intelligent battery management systems.

Keywords Remaining Lithium-ion battery, Remaining Useful Life (RUL), Hybrid CNN-DNN, Binary Particle Swarm Optimization (BPSO), Wrapper feature selection, Deep learning prognostics

Lithium-ion batteries have become the dominant energy-storage technology in electric vehicles, portable electronics, and grid-support applications. However, their gradual degradation—driven by cycling and operating conditions—causes capacity fade, resistance growth, and thermal stress that can lead to performance loss and safety risks. Consequently, accurate battery prognostics, especially RUL estimation, are a core requirement for intelligent Battery Management Systems (BMS) to enable condition-based maintenance, reduce downtime, and prevent unexpected end-of-life failures^{1,2}.

Despite these advances, battery RUL prediction remains challenging for three main reasons. First, degradation signals are highly nonlinear and noisy, and are influenced by temperature and load profiles. Second, many deep architectures rely on complex temporal modeling (e.g., LSTMs, GRUs, and Transformers), which may increase training costs and make them sensitive to data size and hyperparameter choices. Third, multivariate battery datasets frequently contain redundant or weakly informative features, which can degrade regression performance and reduce model interpretability, motivating principled feature selection and optimization. Optimization-assisted learning (including PSO-based schemes) has recently been explored to improve prediction quality, highlighting the importance of embedding optimization within the prognostics pipeline^{3–5}.

From the recent literature, a clear trend is the growth of attention- and transformer-based prognostics and hybrid CNN–RNN models^{6–8}. However, there remains a practical gap for accurate yet lightweight alternatives that (i) effectively learn from multivariate cycle-wise measurements, (ii) reduce feature redundancy via binary

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feature selection, and (iii) provide transparent benchmarking against common deep baselines under the same evaluation protocol. In particular, fewer studies investigate CNN–DNN hybrids (without explicit recurrent blocks) coupled with binary metaheuristic feature selection for RUL regression on cycle-wise degradation datasets, despite their potential advantages in simplicity and robustness.

This study aims to:

1. Develop a hybrid CNN–DNN regression framework for lithium-ion battery RUL prediction that combines convolutional representation learning with deep nonlinear regression.
2. Integrate Binary Particle Swarm Optimization (BPSO) to select an optimal subset of degradation-related features and reduce input redundancy.
3. Provide a consistent experimental protocol and benchmarking against widely used deep baselines (CNN, DNN, LSTM, GRU, BiGRU) using multiple error metrics.

The main contributions of this study are summarized as follows:

- The proposed framework combines convolutional neural networks for automatic feature extraction with deep neural networks for nonlinear regression, enabling effective modeling of complex degradation patterns in lithium-ion batteries.
- A wrapper-based BPSO strategy is employed to identify the most informative subset of battery degradation features, reducing feature redundancy and improving the predictive performance and stability of the model.
- Unlike many recent approaches that rely on complex recurrent or transformer-based architectures, the proposed CNN–DNN model provides a computationally efficient alternative while maintaining high prediction accuracy.
- The proposed model is systematically compared with CNN, DNN, LSTM, GRU, and BiGRU models under the same experimental conditions, demonstrating superior predictive performance across multiple regression metrics.
- The study further investigates the importance of selected features using correlation analysis and SHAP-based explanations, providing insights into how battery health indicators influence RUL prediction.

Battery RUL prediction remains challenging due to the nonlinear degradation behavior of lithium-ion batteries, the presence of redundant or weakly informative features in multivariate datasets, and the trade-off between model complexity and predictive accuracy. The proposed hybrid CNN–DNN framework addresses these challenges by combining convolutional feature extraction with deep nonlinear regression while integrating BPSO-based feature selection to identify the most relevant degradation variables. This joint optimization strategy enables the model to capture complex degradation dynamics, reduce input redundancy, and maintain computational efficiency, thereby improving both prediction accuracy and model robustness.

The remainder of this paper is organized as follows: Sect. 2 mentions the prior studies related to RUL. Section 3 describes the dataset, preprocessing, and the proposed CNN–DNN with BPSO-based feature selection. Section 4 reports and discusses the comparative results. Finally, Sect. 5 concludes the paper and outlines future research directions.

Related works

Recent years have witnessed significant progress in RUL prediction for lithium-ion batteries, driven primarily by advances in data-driven modeling and deep learning. Traditional model-based and statistical approaches, while offering physical interpretability and uncertainty quantification, often struggle to capture the highly nonlinear degradation behavior, capacity regeneration phenomena, and noise-contaminated measurements observed in real-world battery operation. To overcome these limitations, researchers have increasingly adopted machine learning and deep learning models, including convolutional neural networks (CNNs), recurrent architectures such as LSTM and GRU, transformer-based models, and hybrid frameworks that combine signal processing, optimization, and ensemble learning. These methods aim to improve feature representation, temporal dependency modeling, robustness to sparse or noisy data, and generalization across battery chemistries and operating conditions. Moreover, recent studies have explored uncertainty-aware prediction, interval estimation, feature selection via metaheuristic optimization, and adaptive model fusion to enhance reliability and deployment readiness. Despite these advances, challenges remain in balancing prediction accuracy, computational efficiency, data requirements, and model complexity, motivating continued research into lightweight yet effective hybrid architectures for practical battery RUL prognostics.

Model-driven methods

Classical model-based approaches (e.g., electrochemical or stochastic degradation models) can be effective when physics parameters are known, yet they often require strong assumptions and extensive calibration. As a result, data-driven learning has emerged as a practical alternative, leveraging measurements such as voltage, current, and temperature to infer degradation trajectories. Recent studies show strong progress in using deep learning architectures—particularly recurrent models and attention mechanisms—to capture nonlinear aging dynamics. For example, transformer-based and attention-enhanced methods have been developed for battery capacity/RUL prediction, demonstrating improved temporal modeling and early-life forecasting capabilities compared with standard RNN variants^{6,7}. Likewise, hybrid deep models that combine convolutional feature extractors with sequence learners (e.g., CNN–LSTM–Attention with transfer learning) have also reported enhanced stability and generalization across different operating conditions⁸.

Data-driven methods

Traditional machine learning

Li et al.⁹ introduced a genetic algorithm–optimized backpropagation neural network (GA-BP) for predicting the RUL of lithium-ion batteries. This model enhances robustness and minimizes the need for extensive feature engineering by utilizing raw cycle–capacity data. A genetic algorithm optimizes network weights and thresholds, thereby improving stability and accuracy. The framework was validated across various public datasets of different battery chemistries, including LiCoO₂ and LiFePO₄. Experimental results showed that the GA-BP model outperformed traditional BP networks, achieving R^2 values over 0.98 and effectively capturing degradation trends and capacity changes.

Wang et al.¹⁰ proposed an embedded feature selection framework, LS-Embed-BHRSA, for lithium-ion battery RUL prediction. This method combines filtering and wrapper strategies by integrating the Laplace Score with a Binary Hybrid Reptile Search Algorithm for optimal feature selection. Additionally, a stacking-based ensemble learning strategy was employed to enhance RUL prediction by leveraging multiple machine learning models. Evaluated on the Stanford–MIT Battery Life dataset, the framework consistently surpassed traditional machine learning methods in RMSE, MAE, MAPE, and R^2 , demonstrating improved predictive accuracy and robustness in battery prognostics.

Deep learning

Zhang et al.¹¹ developed a hybrid parallel residual convolutional neural network (HPR-CNN) for predicting the RUL of lithium-ion batteries under varying conditions. The model integrates multivariate charging profiles—voltage, current, and temperature—using residual convolutional blocks to identify degradation features. It efficiently utilizes sparse data by leveraging partial charging information (only 20% of total capacity) and inter-cycle differences. The HPR-CNN, supported by a cloud-based system, provides reliable online RUL predictions. Experimental results on a public dataset showed a test error rate of 4.15%, outperforming existing methods and indicating its effectiveness for batteries with random charging patterns.

Zhao et al.¹² developed a hybrid deep learning framework that combines 2D-CNN with a self-attention mechanism to enhance the prediction of battery RUL. The 2D-CNN captures localized degradation features from voltage–capacity data, while the self-attention mechanism focuses on key spatial and temporal regions, boosting accuracy and efficiency. The model utilizes partial-state data segments and a sliding-window approach to minimize redundancy and computation costs. Experiments on datasets with over 240,000 cycles from various battery chemistries, including LFP, NMC, and NCA, demonstrated impressive robustness, low RMSE values, rapid training convergence, and suitability for real-time battery diagnostics and management applications.

Chae et al.¹³ proposed an integrated framework to tackle the challenges of ambient noise and variability in lithium-ion battery State of Health (SoH) estimation and RUL prediction. The framework combines a Denoising Convolutional Neural Network (DnCNN) with a conventional CNN, utilizing residual learning to minimize unknown noise in battery degradation signals. Evaluated on experimental data across diverse operating conditions, the DnCNN–CNN model demonstrated superior accuracy and robustness compared to existing denoising methods, thereby enhancing both SoH estimation and RUL prediction in practical battery management applications.

Liu et al.¹⁴ introduced a RUL prediction framework based on the Mamba architecture, which overcomes traditional sequence modeling shortcomings by addressing long-range dependencies and facilitating parallel computation. The model employs learnable scaling parameters in residual blocks and integrates a multi-head attention mechanism to improve feature representation post-convolution. Data preprocessing included min–max normalization and exponential smoothing to enhance training stability and convergence speed.

Xia et al.¹⁵ introduced an interval-based framework for predicting the RUL of lithium-ion batteries that accommodates varying operational demands and degradation uncertainties. By utilizing fuzzy information granulation and a trend-aware segmentation strategy, the method distinguishes between different degradation behaviors within time-series segments. A health indicator linked to battery capacity is derived from charge–discharge characteristics and is bounded by upper and lower limits. Variational mode decomposition (VMD) is employed to enhance signal quality before predicting with GRU networks. The final RUL interval combines the predicted bounds, enabling uncertainty-aware prognostics.

Fei et al.¹⁶ proposed an attention-assisted temporal convolutional memory-augmented network (ATCMN) for predicting the RUL of lithium-ion batteries using only a few incomplete cycles. This framework can estimate RUL from as few as ten partial discharge cycles at any aging stage by utilizing a three-dimensional tensor representation that combines time, capacity, and temperature information. The ATCMN employs an attention mechanism to prioritize battery parameters and aging cycles, a temporal convolution module to discern spatio-temporal degradation patterns, and a memory-augmented module for historical feature representation. Experiments show that ATCMN surpasses existing methods in prediction accuracy, inference speed, and generalization across battery chemistries and conditions.

Liu et al.¹⁷ proposed a patch-based transformer architecture, PatchFormer, to predict the RUL of lithium-ion batteries, accounting for nonlinear degradation and capacity regeneration. The model features a Dual Patch-wise Attention Network (DPAN) that captures both global dependencies across patches and local temporal dynamics within each patch. Additionally, a Feature-wise Attention Network (FAN) models inter-feature relationships using self-attention on fused temporal representations. This hierarchical attention design integrates global degradation trends with localized regeneration effects in segmented time-series data. Evaluations across three public battery degradation datasets showed that PatchFormer significantly outperforms existing time-series forecasting models, confirming its robustness and accuracy in RUL estimation.

Zhou et al.¹⁸ addressed the high computational cost, decoding inefficiency, and error accumulation issues commonly associated with transformer-based battery RUL prediction models by proposing an optimized

Informer-based forecasting framework. Their approach integrates multi-feature engineering, extracting seven informative battery degradation features and feeding them into the Informer architecture to enhance representation learning while reducing redundancy. To further improve predictive accuracy and computational efficiency, Bayesian optimization is employed for automated hyperparameter tuning. The proposed method was validated using CALCE and NASA battery datasets and demonstrated substantial performance gains, outperforming conventional LSTM models by more than 60% in MAE and standard Transformer models by over 40%, while significantly lowering computational burden.

Hybrid Methods (Model-Data Fusion)

Jafari and Byun¹⁹ developed a CNN–GRU hybrid model that processes voltage, current, temperature, and capacity through parallel CNN branches followed by GRU-based temporal learning, achieving superior RUL prediction accuracy on NASA battery datasets compared with LSTM- and CNN-based baselines.

Thelen et al.²⁰ proposed a hybrid prognostics framework that improves battery RUL predictions by combining model-based and data-driven methods. Their strategy involves offline training of data-driven models to identify prediction errors and online correction of RUL estimates. This approach retains the benefits of interpretability and uncertainty quantification from traditional models while enhancing predictive accuracy. Evaluated on five datasets with 237 battery cells, it showed a 40% reduction in RMSE and a 34% decrease in uncertainty calibration error compared to model-based methods. The framework offers more reliable uncertainty estimates, making it suitable for real-time predictive maintenance and decision-making.

Fan et al.²¹ proposed a hybrid RUL prediction framework for lithium-ion batteries that addresses nonlinear degradation and local capacity regeneration. The method uses Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) to separate battery capacity signals into low- and high-frequency components, thereby improving prediction accuracy. The low-frequency component is modeled using SVR for long-term trends, while the high-frequency component is predicted using an LSTM network trained with the Sparrow Search Algorithm (SSA). Final RUL estimates are derived by combining the outputs of both models. Experimental results on NASA battery datasets indicated high prediction accuracy, with RUL errors within one cycle, RMSE under 0.0086 Ah, MAE under 0.0060 Ah, and R^2 values above 0.96, demonstrating the framework's effectiveness in battery prognostics.

Shuai et al.²² developed a probabilistic framework for predicting the RUL and SOH of lithium-ion batteries by enhancing conventional particle-filter prognostics with grey system theory. They introduced the Grey Model–Particle Flow Filter (GM–PFF) to mitigate particle degradation and uncertainty issues in standard particle flow filters. The framework uses least-squares estimation to initialize a double-exponential empirical degradation model and employs a grey model to obtain accurate observation values. This approach refines model parameters to produce probabilistic RUL estimates during different battery ageing stages. Experimental results on CALCE and NASA PCoE datasets demonstrated that GM–PFF improves prediction accuracy, reducing RMSE by about 1% compared to standard PFF, and provides better uncertainty quantification.

Dong et al.²³ proposed a hybrid framework for predicting the RUL of lithium-ion batteries that addresses the challenges of limited historical data and measurement noise. Their model combines advanced data preprocessing with a CNN–Least Squares Support Vector Regression to enhance degradation modeling accuracy. They introduce novel RUL indices and health factors based on battery capacity and resistance. The method incorporates Multi-Resolution Singular Value Decomposition for noise suppression and utilizes a mutant Particle Swarm Optimization algorithm for hyperparameter optimization. Experimental results indicate significant improvements over traditional methods, with reductions of at least 37% in MAE and 61% in RMSE, showcasing the benefits of noise reduction and hybrid learning for accurate RUL prediction under challenging conditions.

Li et al.²⁴ presented a new approach to battery RUL prediction that overcomes the limitations of conventional methods, which only account for a consistent degradation trend. They introduced a hybrid grey forecasting framework that models both normal degradation and capacity regeneration in lithium-ion batteries. To enhance long-term prediction accuracy and reduce error accumulation, they incorporated an Ensemble Kalman Filter (EnKF) to dynamically adjust predictions. Validation through ablation studies confirmed the effectiveness of the hybrid approach, which outperformed data-driven methods, achieving a mean absolute percentage error of less than 1%, thus accurately reflecting complex battery degradation behaviors.

Safavi et al.²⁵ proposed a hybrid data-driven framework for early RUL prediction of lithium-ion batteries by integrating deep feature extraction and ensemble learning. Their approach combines a CNN with XGBoost, where the CNN automatically extracts degradation-related features from discharge capacity profiles using image-based representations. These learned features are further enriched with statistical and operational features derived from the first 100 battery cycles, incorporating charging policy information. To enhance model performance, the Coati Optimization Method (COM) is employed to optimally tune the CNN hyperparameters. Experimental results demonstrated that the proposed CNN–XGBoost model achieves reliable early-life RUL prediction, attaining an RMSE of 106 cycles and a MAPE of 7.5%, confirming the effectiveness of integrating deep feature learning, ensemble regression, and metaheuristic optimization for battery prognostics.

Zhao et al.²⁶ proposed a fusion neural network combining a Broad Learning System (BLS) and an LSTM network to enhance lithium-ion battery capacity and RUL prediction. The BLS extracts features from historical data, which are then fed into the LSTM for effective temporal modeling, thereby circumventing the need for deep architectures. Validation with NASA and CALCE datasets showed that the BLS–LSTM model maintained accurate predictions even with a 25% reduction in training dataset size, demonstrating its data efficiency and computational effectiveness.

Yao et al.²⁷ developed a hybrid deep learning framework for predicting the RUL of lithium-ion batteries, with a focus on safety-critical applications. Their approach combines statistical and physical degradation features

using an LSTM–CNN architecture. Statistical features are obtained from Gaussian mixture models applied to incremental capacity data, while physical features are based on discharge cycle characteristics. This integrated feature set is processed and used as input for the LSTM–CNN network, enabling it to capture both temporal dependencies and nonlinear degradation patterns. Testing on the NASA battery dataset showed substantial improvements over traditional LSTM models, achieving reductions of 31.0% in MAE, 29.4% in RMSE, and 31.4% in MAPE, while keeping average RUL prediction errors within one cycle, validating the framework's effectiveness for battery prognostics.

Wu et al.²⁸ introduced a framework for predicting the RUL of lithium-ion batteries, addressing challenges from energy regeneration and localized degradation. Their method employs a denoising autoencoder for signal cleaning, followed by a one-dimensional CNN for feature extraction. Temporal dependencies are captured using a bidirectional GRU (BiGRU) network, which includes a quantile regression layer for uncertainty quantification in predictions. The model, trained with a quantile regression loss, enhances robustness and reliability. Tests on NASA and CALCE datasets show that this CNN–BiGRU–QR framework surpasses traditional GRU approaches, demonstrating its efficacy in RUL prediction.

Wang et al.²⁹ explored the prediction of internal temperature in lithium-ion batteries under dynamic conditions, addressing challenges posed by temperature fluctuations and response delays. They proposed an enhanced deep learning model combining Convolutional Neural Networks (CNN), Bidirectional Long Short-Term Memory (Bi-LSTM), and an attention mechanism (CNN–Bi-LSTM–AM). This model extracts spatial correlations and captures temporal dependencies while highlighting the critical features that affect temperature changes. Using simulation data, the model demonstrated improved prediction accuracy of 1.2–2.3% over traditional methods such as ARIMA and standard LSTM, with minimal impact on inference time.

Du et al.³⁰ introduced a Deep Reinforcement Learning Fusion Network (DRLFN) to overcome the shortcomings of single-model approaches in estimating the Remaining Useful Life (RUL) of lithium-ion batteries. The framework uses LSTM and Transformer models to make initial RUL predictions, which are then integrated and enhanced through a Deep Q-Network (DQN). To ensure robust fusion, the method distinguishes between regular and irregular prediction areas, identifies bias patterns in regular regions, and optimizes model weights using DQN. In contrast, irregular regions utilize credibility scores from individual predictions to dynamically choose the best model. Results from experiments on the MIT–Stanford–Toyota lithium battery dataset show that DRLFN outperforms traditional models, demonstrating the effectiveness of reinforcement learning-driven adaptive fusion in precise RUL predictions.

Recent advances in battery prognostics and time-series prediction have highlighted the effectiveness of hybrid deep learning models that combine convolutional neural networks with other neural architectures. Convolutional layers are widely used for automatic feature extraction from multivariate signals, while dense or sequential neural layers capture complex nonlinear relationships and temporal dependencies. For instance, hybrid CNN–LSTM and CNN–BiLSTM models have been proposed for lithium-ion battery Remaining Useful Life prediction, where CNN layers learn degradation-related features from voltage, current, and temperature signals, and recurrent layers capture temporal evolution across charging cycles^{31,32}.

Beyond battery prognostics, hybrid CNN-based deep learning models have also been successfully applied to other predictive tasks such as energy forecasting and multivariate time-series modeling, where they demonstrate improved learning of complex patterns and reduced prediction errors compared with standalone models³³.

Although existing studies have achieved promising results, several challenges remain in balancing model complexity, computational efficiency, and predictive accuracy. Many deep learning models rely on complex architectures that may be difficult to deploy in practical battery management systems. In addition, redundant input features can negatively affect model performance when feature selection is not properly considered. Motivated by these challenges, the present study proposes a hybrid CNN–DNN framework combined with BPSO-based feature selection, which aims to improve prediction accuracy while maintaining a relatively lightweight and computationally efficient model structure.

Table 1 provides a comprehensive comparison of recent representative studies on lithium-ion battery RUL prediction, highlighting diverse modeling paradigms ranging from hybrid deep learning architectures and signal decomposition techniques to probabilistic and attention-based frameworks.

In summary, existing studies on lithium-ion battery RUL prediction demonstrate substantial progress through deep learning, hybrid modeling, optimization-based feature selection, uncertainty-aware forecasting, and ensemble fusion strategies. While attention-based, transformer-driven, probabilistic, and signal-decomposition-assisted models have achieved impressive predictive accuracy, many of these approaches rely on complex architectures, extensive data requirements, or computationally intensive optimization and fusion mechanisms, which may limit their scalability and real-time applicability in practical battery management systems. Moreover, several methods focus primarily on temporal modeling or uncertainty quantification without explicitly addressing feature redundancy and model simplicity. These limitations highlight the need for a robust yet lightweight framework that can effectively exploit multivariate battery data while maintaining high accuracy and computational efficiency. Motivated by these gaps, the present study proposes an optimized hybrid CNN–DNN model enhanced with Binary Particle Swarm Optimization for feature selection, aiming to balance predictive performance, model complexity, and practical deployability for reliable battery RUL prognostics.

Materials and methods

Dataset description

This study utilizes a publicly available lithium-ion battery degradation dataset obtained from the Kaggle repository³⁴, which provides cycle-level operational and health measurements for lithium-ion batteries under controlled aging conditions. The dataset is designed to support research on battery health assessment and RUL prediction, making it suitable for evaluating data-driven prognostic models.

Ref.	Main Objective	Dataset	Core Methodology	Key Techniques	Main Findings/Limitations
25	Early RUL prediction	Public battery dataset	CNN-XGBoost	Image-based CNN features; COM hyperparameter tuning	Achieved RMSE = 106 cycles, MAPE = 7.5%; relies on handcrafted cycle-window design
19	RUL prediction	NASA	CNN-GRU	Parallel CNN branches + GRU temporal modeling	Outperformed LSTM/CNN; limited feature optimization
20	RUL with uncertainty	Multiple datasets (237 cells)	Hybrid model-based + data-driven	Error correction, uncertainty propagation	40% RMSE reduction; higher complexity
11	Online RUL prediction	Public dataset	HPR-CNN	Residual CNN; sparse charging data	Test error 4.15%; depends on cloud infrastructure
16	RUL from limited cycles	Public datasets	ATCMN	Attention + temporal convolution + memory	High accuracy with 10 cycles; complex architecture
26	Capacity & RUL prediction	NASA, CALCE	BLS-LSTM	Broad learning + temporal modeling	Accurate with 25% data; limited nonlinear feature learning
12	RUL across chemistries	Large multi-chemistry datasets	2D-CNN + Attention	Sliding window; partial SOC	Low RMSE; high computational cost
21	RUL with regeneration	NASA	CEEMDAN-SVR-LSTM	Signal decomposition + SSA	RUL error ≤ 1 cycle; heavy preprocessing
10	RUL with feature selection	Stanford-MIT	LS-Embed-BHRSA + Stacking	Filter-wrapper FS + ensemble	Improved RMSE/MAE; model complexity
22	Probabilistic RUL	NASA, CALCE	GM-PFF	Grey model + particle filtering	Better uncertainty handling; marginal RMSE gain
17	RUL with regeneration	Public datasets	PatchFormer	Dual patch-wise & feature-wise attention	Strong accuracy; transformer complexity
13	Noise-robust RUL & SOH	Experimental data	DnCNN-CNN	Residual denoising	Improved robustness; noise-focused
14	General RUL prediction	Public datasets	Mamba-based model	Attention + residual scaling	Handles long sequences; limited feature selection
15	Interval RUL prediction	Public datasets	Fuzzy-VMD-GRU	Granulation + uncertainty bounds	Interval prediction; not point-optimized
23	RUL with noise & small data	Public datasets	CNN-LSSVR + MPPO	MRSVD denoising + hybrid learning	MAE $\downarrow$ 37%, RMSE $\downarrow$ 61%; multiple stages
27	Safety-oriented RUL	NASA	LSTM-CNN	Statistical + physical features	MAE $\downarrow$ 31%; relies on IC curve extraction
28	Uncertainty-aware RUL	NASA, CALCE	DAE-CNN-BiGRU-QR	Quantile regression	Improved robustness; higher inference cost
29	Temperature prediction	Simulated data	CNN-BiLSTM-Attention	Thermal modeling	1.2–2.3% accuracy gain; indirect RUL
9	RUL without FS	Multi-chemistry datasets	GA-BP	GA weight optimization	$R^2 > 0.98$; lacks deep feature extraction
30	Adaptive RUL fusion	MIT-Stanford	DRLFN	LSTM + Transformer + DQN	Improved accuracy; complex fusion logic
24	RUL with regeneration	Public datasets	Grey-EnKF	Grey modeling + filtering	MAPE $< 1\%$; limited scalability
18	Efficient transformer RUL	NASA, CALCE	Informer + BO	Feature engineering + BO	MAE $\downarrow$ 60%; still transformer-heavy

Table 1. Summary of Representative Related Works on Lithium-Ion Battery RUL Prediction.

The dataset contains measurements from multiple battery cells, capturing electrical, thermal, and degradation-related parameters recorded throughout the charge-discharge cycling process. Each record corresponds to a specific battery cycle and includes both input features and target health indicators, enabling supervised regression modeling.

The dataset consists of 680 samples, each representing a single operational cycle of a battery. The data include measurements of charging and discharging behavior, thermal characteristics, battery capacity, state of health, and remaining useful life.

The target variable of interest in this study is the RUL, defined as the number of cycles remaining before the battery reaches its end-of-life threshold.

Table 2 presents a descriptive statistical analysis of the lithium-ion battery dataset used in this study, summarizing the distributions and variabilities of all input features and target variables across 680 samples. The statistics indicate a wide range of operating cycles (1–250), confirming that the dataset captures both early-life and near-end-of-life battery behavior, which is essential for reliable RUL modeling. Electrical and thermal parameters during charging and discharging (chI, chV, chT, disI, disV, disT) exhibit moderate standard deviations, reflecting realistic variations in operating conditions without extreme outliers. The battery capacity (BCt) and State of Health (SOH) show clear degradation trends, with SOH decreasing from nearly 100% to approximately 37%, indicating substantial aging coverage. Moreover, the RUL spans from 0 to 249 cycles, with a mean value of approximately 113 cycles, highlighting the suitability of the dataset for supervised regression-based RUL prediction.

Figure 1 illustrates the Pearson correlation heatmap among the battery features and target variables, providing insight into the linear relationships within the dataset. A strong positive correlation is observed among BCt, SOH, and RUL, indicating that higher battery capacity and state of health are directly associated with longer remaining useful life, consistent with battery degradation theory. Conversely, the cycle feature shows a strong negative correlation with BCt, SOH, and RUL, indicating that battery aging progresses with increasing cycles. Most charging and discharging electrical and thermal variables (chI, chV, chT, disI, disV, disT) show weak to moderate correlations with RUL, suggesting that their influence on degradation is nonlinear and interdependent rather than purely linear.

Figure 2 presents the kernel density estimation (KDE) plots for all battery-related features and target variables, illustrating their empirical probability distributions across the dataset. The distributions of charging

	count	mean	std	min	50%	max
battery_id	680.0	1.044118	0.830810	0.000000	1.000000	2.000000
cycle	680.0	114.470588	66.601594	1.000000	114.000000	250.000000
chI	680.0	1.400069	0.102903	1.000859	1.401904	1.747515
chV	680.0	4.202765	0.046973	4.035078	4.202503	4.359233
chT	680.0	26.871113	1.489824	21.604784	26.823350	30.913077
disI	680.0	2.007630	0.097797	1.702438	2.006990	2.411226
disV	680.0	3.501070	0.304160	2.484857	3.496967	4.363548
disT	680.0	33.099894	1.827415	26.848457	33.105426	38.388721
BCt	680.0	1.427851	0.333250	0.750972	1.429413	1.993692
SOH	680.0	71.392565	16.662488	37.548593	71.470656	99.684587
RUL	680.0	113.470588	66.601594	0.000000	113.000000	249.000000

Table 2. Descriptive statistical summary of the lithium-ion battery dataset used for RUL prediction.

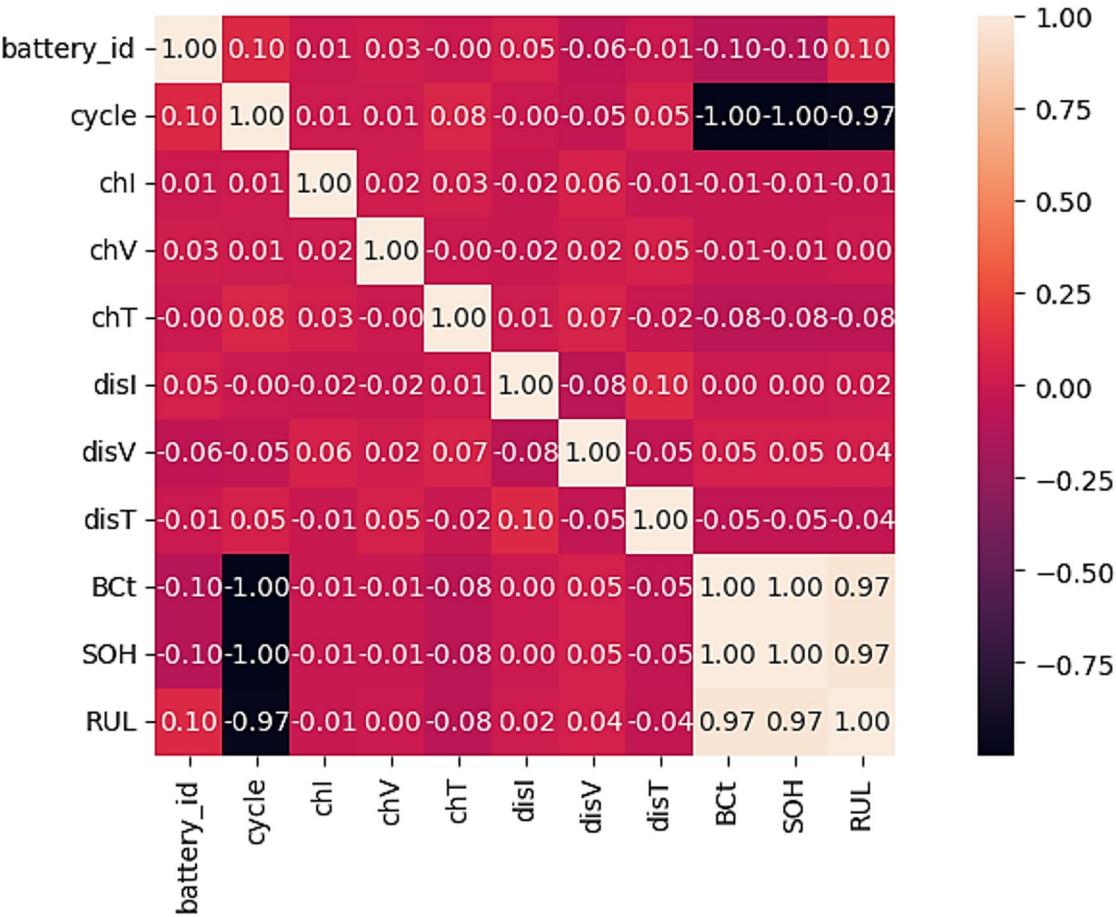


Fig. 1. Pearson correlation heatmap of battery operational features, State of Health (SOH), and Remaining Useful Life (RUL).

and discharging electrical parameters (chI, chV, disI, disV) and temperature variables (chT, disT) exhibit approximately unimodal, near-Gaussian shapes, indicating stable operating conditions with limited extreme values. The BCt and SOH distributions span a wide range, reflecting progressive battery degradation from healthy to aged states. Similarly, the RUL distribution covers early, mid, and late life stages, confirming that the dataset provides comprehensive degradation coverage suitable for supervised learning. The smooth, continuous density profiles further indicate the absence of severe skewness or abnormal outliers, supporting the dataset's reliability and justifying its use for training deep learning models for accurate RUL prediction.

Figure 3 illustrates the pairwise scatter plot matrix of all battery operational features and target variables, providing a comprehensive view of their bivariate relationships and marginal distributions. The diagonal

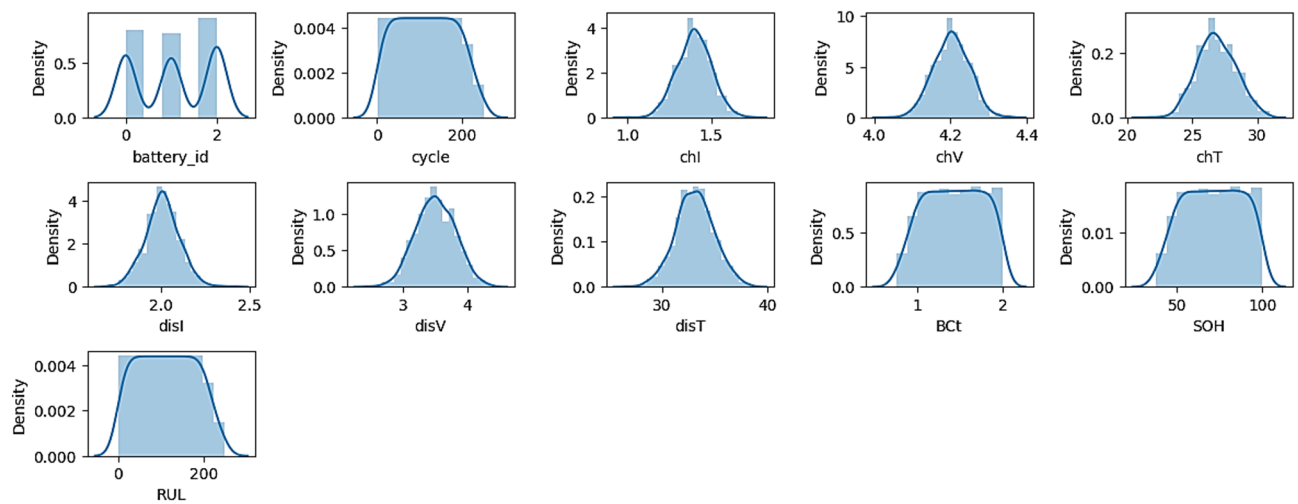


Fig. 2. Kernel density estimation (KDE) plots showing the distribution of battery operational features, SOH, and RUL.

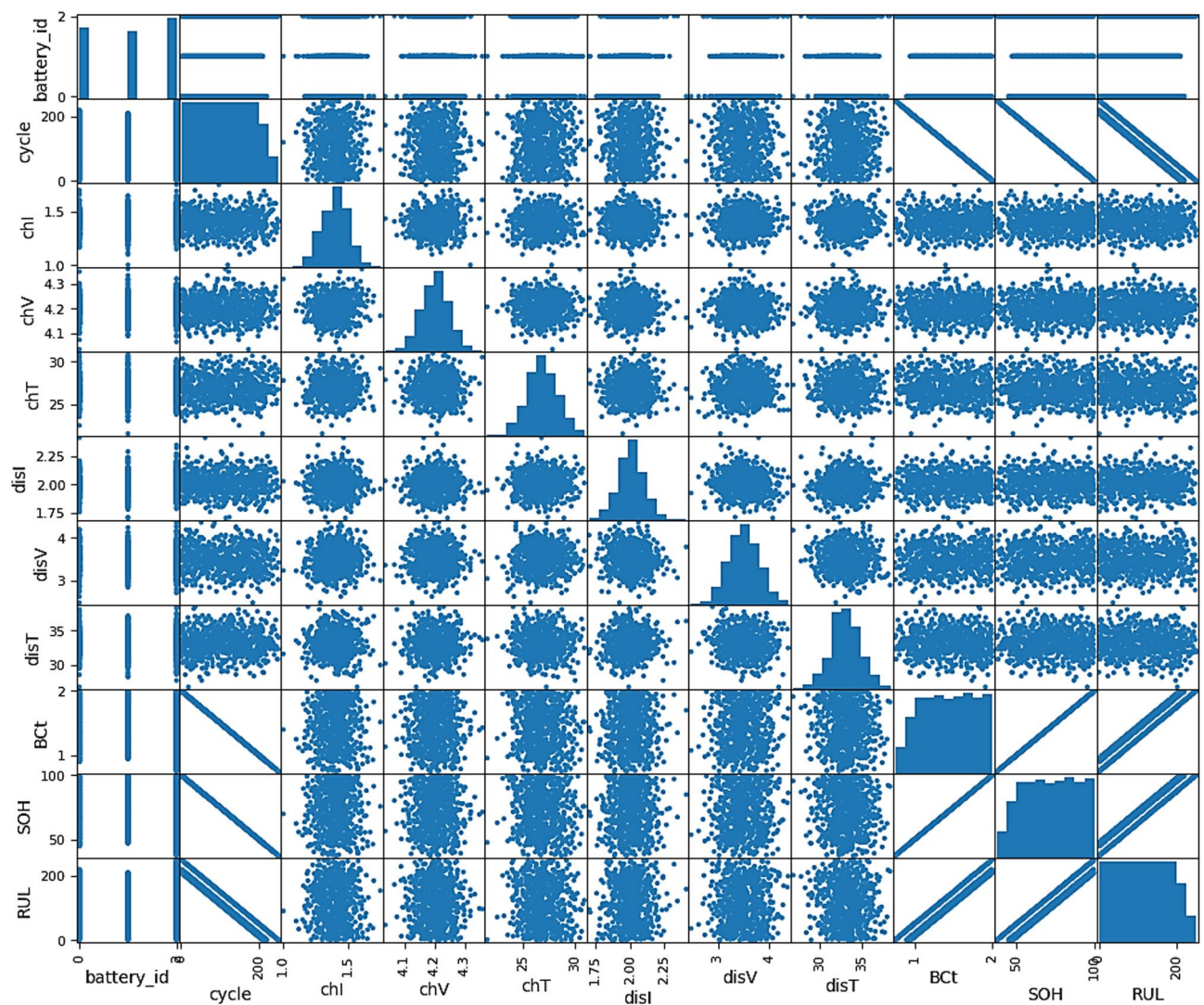


Fig. 3. Pairwise scatter plot matrix showing relationships among battery operational features, SOH, and RUL.

histograms show that most electrical and thermal variables follow approximately unimodal distributions, while off-diagonal scatter plots reveal complex, nonlinear relationships among features. A clear monotonic degradation pattern is observed between cycle and the degradation indicators (BCt, SOH, and RUL), where increasing cycle count corresponds to decreasing battery capacity and remaining life. In contrast, charging and discharging current, voltage, and temperature features exhibit dispersed and overlapping patterns, indicating weak linear correlations and reinforcing the need for nonlinear learning models. The dense, well-distributed scatter points also suggest the absence of severe outliers and confirm that the dataset spans a wide range of operating and degradation states.

Figure 4 presents boxplots for all battery operational features and target variables, illustrating their central tendency, dispersion, and presence of potential outliers. The boxplots show that most electrical and thermal parameters (chI, chV, chT, disI, disV, disT) exhibit relatively compact interquartile ranges, indicating stable operating conditions with limited cycle-to-cycle variability. The cycle, SOH, and RUL variables exhibit wider spreads, reflecting the progressive aging of the battery and the natural variability in battery degradation over time. A limited number of outliers are observed in some features, particularly in current and temperature measurements; however, these values remain within physically plausible ranges and do not indicate severe anomalies. The BCt and SOH boxplots further confirm gradual capacity and health decline patterns, while the broad distribution of RUL highlights the inclusion of early-, mid-, and late-life battery states. Overall, the boxplot analysis confirms the robustness and quality of the dataset and supports its suitability for training deep learning models without extensive outlier removal.

Figure 5 illustrates the histogram distributions of all battery operational features and target variables, providing a clear overview of the dataset's frequency and spread. The histograms show that charging and discharging electrical parameters (chI, chV, disI, disV) and temperature-related features (chT, disT) are approximately bell-shaped and centered around their mean values, indicating stable measurement conditions and consistent operating regimes. The cycle distribution spans the full operational lifetime of the batteries, while SOH and BCt exhibit broad ranges that reflect progressive capacity degradation from healthy to aged states. Similarly, the RUL histogram covers early, mid, and late life stages, confirming that the dataset provides comprehensive degradation coverage for supervised learning.

Data preprocessing

Effective data preprocessing is a critical step for improving model convergence, stability, and predictive accuracy in battery RUL estimation. In this study, a systematic preprocessing pipeline was applied to the raw lithium-ion battery dataset before feature selection and model training. The preprocessing stages include data cleaning, normalization, feature–target separation, and dataset partitioning.

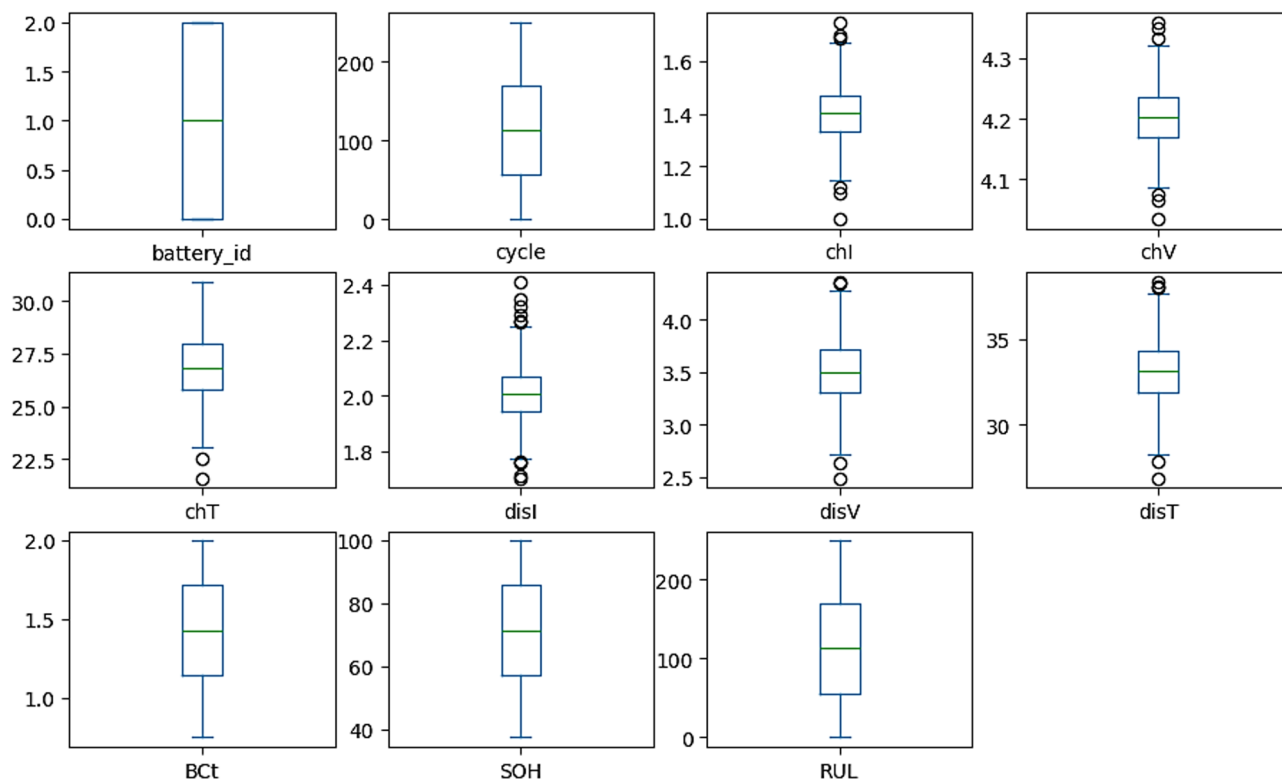


Fig. 4. Boxplot visualization of battery operational features, SOH, and RUL, showing data dispersion and outlier distribution.

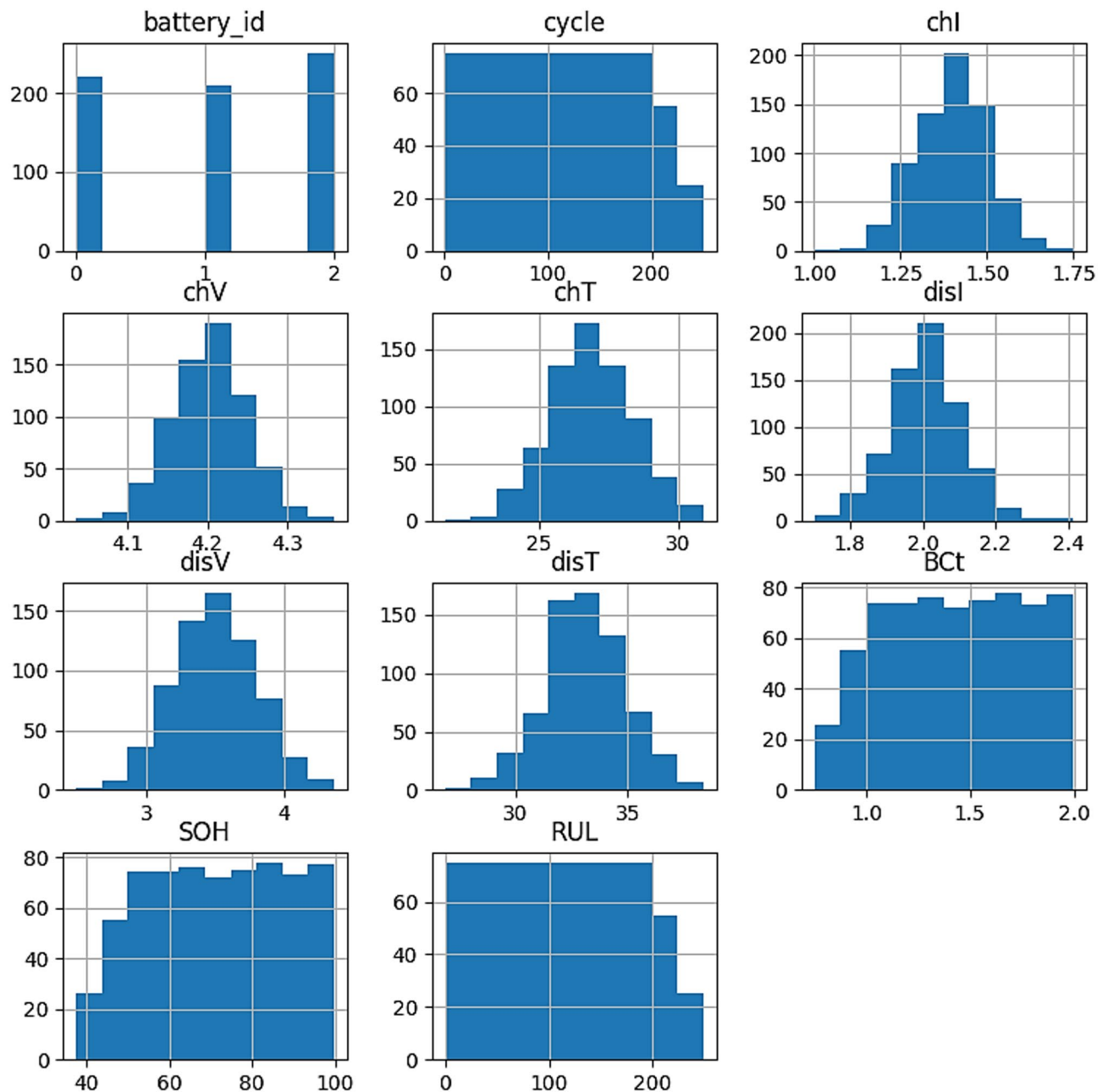


Fig. 5. Histogram distributions of battery operational features, SOH, and RUL in the lithium-ion battery dataset.

Data cleaning and consistency check

The dataset was first examined for missing values, duplicated records, and physically implausible measurements. Since the dataset was generated under controlled experimental conditions, no missing entries were observed. Outlier inspection was performed using statistical visualization (boxplots and density analysis), confirming that extreme values remained within physically meaningful operating ranges. Therefore, no aggressive outlier removal was applied to avoid information loss related to battery degradation behavior.

Feature-Target Definition

Let the dataset be represented as:

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N \quad (1)$$

where $N = 680$ denotes the total number of samples,

$$\mathbf{x}_i = [x_{i1}, x_{i2}, \dots, x_{id}] \in \mathbb{R}^d \quad (2)$$

represents the input feature vector consisting of electrical, thermal, and degradation-related variables, and

$$y_i \in \mathbb{R}$$

denotes the target variable, RUL.

Feature normalization

Due to the heterogeneous scales of battery features (e.g., voltage, current, temperature, and cycle count), normalization was applied to ensure numerical stability and prevent dominant features from biasing the learning process. Min–Max normalization was adopted to rescale all input features into the range $[0, 1]$, defined as:

$$x_{ij}^{\text{norm}} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (3)$$

where: x_{ij} is the original value of the feature j for sample i , $\min(x_j)$ and $\max(x_j)$ denote the minimum and maximum values of feature j , respectively.

This transformation preserves the relative distribution of each feature while facilitating faster convergence of gradient-based learning algorithms.

Target scaling

To maintain consistency between input and output magnitudes during training, the RUL target variable was also normalized using Min–Max scaling:

$$y_i^{\text{norm}} = \frac{y_i - \min(y)}{\max(y) - \min(y)} \quad (4)$$

The predicted RUL values were later transformed back to their original scale during post-processing for evaluation and interpretation.

Feature selection preparation

After normalization, the feature matrix was prepared for BPSO-based feature selection. Let the normalized feature matrix be:

$$\mathbf{X} = [\mathbf{x}_1^{\text{norm}}, \mathbf{x}_2^{\text{norm}}, \dots, \mathbf{x}_N^{\text{norm}}]^T \quad (5)$$

BPSO operates on a binary selection vector:

$$\mathbf{b} = [b_1, b_2, \dots, b_d], b_j \in \{0, 1\} \quad (6)$$

where $b_j = 1$ indicates that the j -th feature is selected, and $b_j = 0$ indicates exclusion. Only selected features are passed to the CNN–DNN prediction model.

Dataset partitioning

To ensure unbiased evaluation and prevent data leakage, the dataset was divided into training, validation, and testing subsets using a 70%–15%–15% split, defined as:

$$\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}} \quad (7)$$

where:

- $|\mathcal{D}_{\text{train}}| = 0.7N$, $|\mathcal{D}_{\text{val}}| = 0.15N$, and $|\mathcal{D}_{\text{test}}| = 0.15N$.

The training set was used to learn model parameters, the validation set for hyperparameter tuning and early stopping, and the test set exclusively for final performance assessment.

While the choice of 70%–15%–15% random split has been used in the current work, which randomly splits the data into training, validation, and testing sets, with balanced proportions of degradation stages, it must be noted that the use of cycle-based battery data could lead to potential information leakage, considering the possibility of having the same battery cell included in more than one set. However, in real-world battery prognostic studies, instead of the conventional method of data splitting, battery-wise data splitting, i.e., the use of leave one battery out, or chronological data splitting based on the sequence of cycles, could be more appropriate, which would ensure the testing of the model with data not seen during the training process.

It is important to note that the lithium-ion battery dataset contains cycle-level observations associated with multiple battery cells, identified through the `battery_id` variable. In the current experimental setup, the dataset was divided using a random 70%–15%–15% split, which ensures balanced coverage of different degradation stages across training, validation, and testing subsets. Under this partitioning strategy, cycles originating from the same battery cell may appear in both the training and testing sets. While this approach is commonly used in data-driven battery prognostics to maintain sufficient training samples, it may introduce potential information overlap when degradation trajectories from the same battery are partially observed during training.

Feature selection using BSO

To select the most informative subset of battery features and reduce redundancy, this study employs BPSO as a wrapper-based feature selection technique. In BPSO, each particle represents a candidate feature subset encoded as a binary vector, and the swarm iteratively searches for the subset that optimizes a defined fitness function.

Particle representation

Let d be the total number of available features.

The i -th particle is represented by:

- Binary position vector (feature mask):

$$\mathbf{x}_i(t) = [x_{i1}(t), x_{i2}(t), \dots, x_{id}(t)], x_{ij}(t) \in \{0, 1\} \quad (8)$$

where $x_{ij}(t) = 1$ indicates selecting feature j , and 0 indicates excluding it.

- Real-valued velocity vector:

$$\mathbf{v}_i(t) = [v_{i1}(t), v_{i2}(t), \dots, v_{id}(t)], v_{ij}(t) \in \mathbb{R} \quad (9)$$

Each particle stores:

- its personal best position $\mathbf{p}_i(t)$, the global best position $\mathbf{g}(t)$ among all particles.

BPSO Update Equations

1) Velocity Update

At iteration t , the velocity of the particle i for dimension j is updated by:

$$v_{ij}(t+1) = \omega v_{ij}(t) + c_1 r_{1,ij}(t) (p_{ij}(t) - x_{ij}(t)) + c_2 r_{2,ij}(t) (g_j(t) - x_{ij}(t)) \quad (10)$$

where: ω is the inertia weight controlling exploration/exploitation, c_1, c_2 are cognitive and social acceleration coefficients, $r_{1,ij}(t), r_{2,ij}(t) \sim U(0, 1)$ are random values.

To avoid unstable oscillations, velocities are often clamped:

$$v_{ij}(t+1) \leftarrow \max(-V_{\max}, \min(v_{ij}(t+1), V_{\max})) \quad (11)$$

2) Transfer Function (Sigmoid)

In binary PSO, the velocity is transformed into a probability of selecting a feature using the sigmoid function:

$$S(v_{ij}(t+1)) = \frac{1}{1 + \exp(-v_{ij}(t+1))} \quad (12)$$

where $S(\cdot) \in (0, 1)$.

3) Binary Position Update

The new binary position is sampled probabilistically:

$$x_{ij}(t+1) = \begin{cases} 1, & \text{if } r_{ij}(t) < S(v_{ij}(t+1)) \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

where $r_{ij}(t) \sim U(0, 1)$.

Thus, higher velocity increases the probability that the feature is selected.

Fitness Function Design (Wrapper Selection)

The goal is to select a subset of features that yields high prediction accuracy while remaining compact.

Let:

- $\mathcal{F}(\mathbf{x}_i)$ be the selected feature subset from mask $\mathbf{x}_i$, $k_i = \sum_{j=1}^d x_{ij}$ be the number of selected features, d be the total features.

We define a multi-objective fitness as a weighted sum:

$$J(\mathbf{x}_i) = \alpha \mathcal{L}(\mathcal{F}(\mathbf{x}_i)) + (1 - \alpha) \left(\frac{k_i}{d} \right) \quad (14)$$

where:

- $\mathcal{L}(\cdot)$ is a prediction loss computed using the learner (here, the CNN-DNN regressor) on the validation set, $\alpha \in [0, 1]$ balances accuracy vs. subset size.

A common choice is using MSE as the loss:

$$\mathcal{L} = \text{MSE} = \frac{1}{n} \sum_{m=1}^n (y_m - \hat{y}_m)^2 \tag{15}$$

The optimization objective is:

$$\min_{\mathbf{x}_i \in \{0,1\}^d} J(\mathbf{x}_i) \tag{16}$$

In the proposed framework, the BPSO algorithm operates as a wrapper-based feature selection mechanism tightly coupled with the CNN–DNN prediction model. Each particle in the swarm represents a candidate subset of input features encoded as a binary vector, where a value of 1 indicates that the corresponding feature is selected and 0 indicates exclusion. For each candidate feature subset generated by a particle, the dataset is reduced accordingly and used as input to train the CNN–DNN regression model. The prediction performance of the CNN–DNN model on the validation dataset is then evaluated using the Mean Squared Error, which serves as the fitness value guiding the BPSO search process. Through iterative updates of particle velocities and positions, the algorithm explores different combinations of features and gradually converges toward the subset that minimizes prediction error while maintaining a compact representation of the input space.

Personal/Global Best Updates
After evaluating fitness $J(\mathbf{x}_i(t+1))$:

- **Personal best:**

$$\mathbf{p}_i(t+1) = \begin{cases} \mathbf{x}_i(t+1), & \text{if } J(\mathbf{x}_i(t+1)) < J(\mathbf{p}_i(t)) \\ \mathbf{p}_i(t), & \text{otherwise} \end{cases} \tag{17}$$

- **Global best:**

$$\mathbf{g}(t+1) = \arg \min_{\mathbf{p}_i(t+1)} J(\mathbf{p}_i(t+1)) \tag{18}$$

Stopping Criteria

The algorithm stops when either:

A maximum number of iterations T is reached: $t \geq T$ Or no improvement occurs for L consecutive iterations (early stopping).

Table 3 summarizes the parameter configuration of the Binary Particle Swarm Optimization algorithm used for feature selection in this study. The swarm size was set to 30 particles, and the optimization process was executed for a maximum of 50 iterations. The inertia weight was set to 0.7 to balance exploration and exploitation during the search process, while the cognitive and social acceleration coefficients were both set to 1.5 to regulate the influence of personal and global best solutions. Velocity values were constrained within a predefined range to ensure stable convergence. Additionally, an early stopping mechanism was employed to terminate the optimization process if no improvement in the global best solution was observed for ten consecutive iterations, thereby improving computational efficiency.

Feature selection analysis

Figure 6 illustrates the Pearson correlation heatmap of the selected feature subset after applying the BPSO algorithm. The results show that BPSO effectively retains the most informative degradation-related variables, namely SOH, Battery Capacity (BCt), and cycle, which exhibit strong correlations and capture the dominant aging dynamics of the battery. At the same time, the algorithm reduces redundancy by excluding weakly informative or highly correlated variables, while preserving complementary electrical features such as charging temperature (chT) and discharging current and voltage (disI, disV) that provide additional operational context.

Parameter	Description	Value
Swarm size	Number of particles in the population	30
Maximum iterations	Total optimization iterations	50
Inertia weight (w)	Controls exploration and exploitation balance	0.7
Cognitive coefficient (c ₁)	Particle self-learning factor	1.5
Social coefficient (c ₂)	Swarm learning factor	1.5
Velocity limits	Maximum absolute velocity value	[−4, 4]
Transfer function	Sigmoid function for binary mapping	Sigmoid
Fitness function	Weighted combination of validation MSE and feature subset size	Equation (6)
Early stopping criterion	Stop if no improvement in global best for 10 consecutive iterations	Enabled

Table 3. Parameter configuration of the Binary Particle Swarm Optimization (BPSO) algorithm used for feature selection.

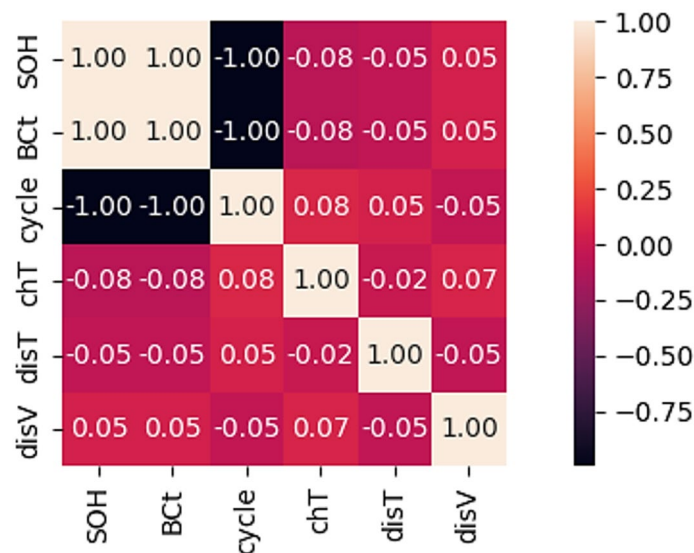


Fig. 6. Correlation heatmap of the selected features after BPSO-based feature selection.

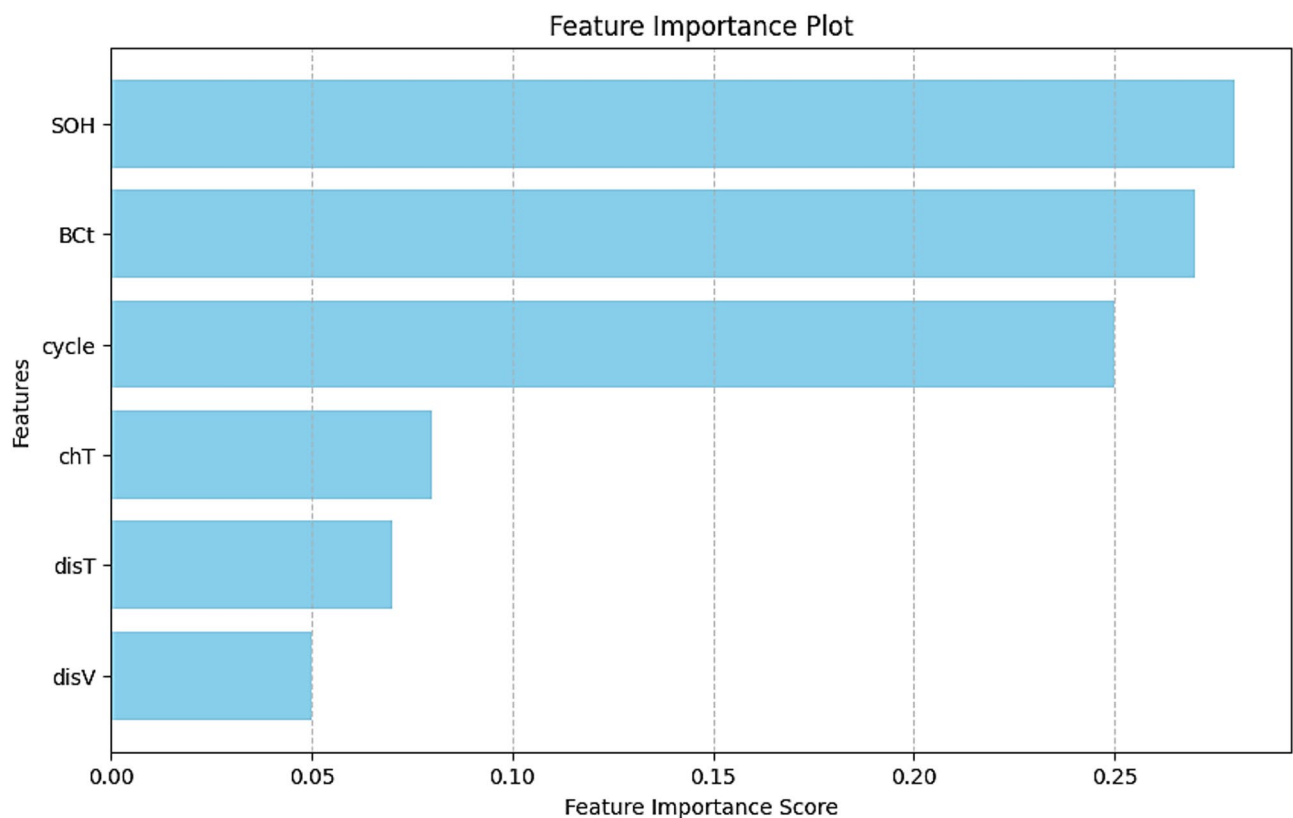


Fig. 7. Feature importance ranking derived from the BPSO-selected feature subset used in the proposed CNN-DNN model for lithium-ion battery RUL prediction.

Figure 7 presents the feature importance analysis of the variables selected by the BPSO algorithm. The results indicate that State of Health (SOH) and Battery Capacity (BCt) are the most influential features for RUL prediction, highlighting their direct and dominant relationship with battery degradation dynamics. The cycle feature also exhibits a high importance score, confirming that cumulative usage history remains a critical indicator of battery aging. In contrast, thermal and electrical features such as charging temperature (chT), discharging temperature (disT), and discharging voltage (disV) contribute supplementary information, capturing operational conditions that indirectly affect degradation behavior. The relative ranking shows that BPSO successfully prioritizes

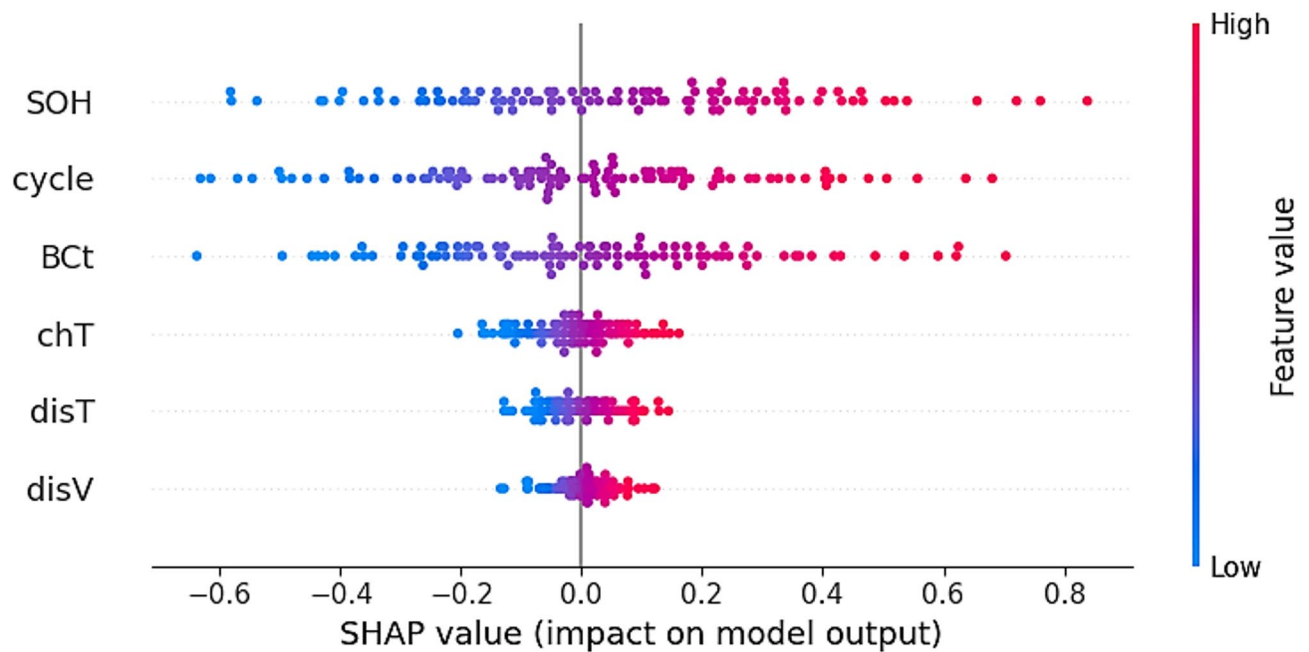


Fig. 8. SHAP summary plot illustrating the impact of selected features on the CNN-DNN model's RUL predictions after BPSO-based feature selection.

1. **Input:** Feature matrix X , target y , swarm size N_p , iterations T , ω , c_1 , c_2 , α
 2. Initialize particles $\mathbf{x}_i(0) \in \{0,1\}^d$ and velocities $\mathbf{v}_i(0) \in \mathbb{R}^d$ randomly
 3. For each particle, compute $J(\mathbf{x}_i(0))$; set $\mathbf{p}_i(0) = \mathbf{x}_i(0)$
 4. Set global best $\mathbf{g}(0)$ as the best $\mathbf{p}_i(0)$
 5. **For** $t = 0$ to $T - 1$:
 • Update velocities using the velocity equation
 • Compute probabilities using sigmoid $S(\cdot)$
 • Update binary positions using probabilistic rule
 • Evaluate fitness $J(\mathbf{x}_i(t+1))$ using CNN-DNN validation loss and feature ratio
 • Update $\mathbf{p}_i(t+1)$ and $\mathbf{g}(t+1)$
 6. **Output:** Selected feature subset given by $\mathbf{g}(T)$

Algorithm 1: BPSO for Feature Selection.

physically meaningful, degradation-sensitive features while retaining complementary variables that enhance model robustness. Overall, this analysis validates the effectiveness of the proposed feature selection strategy in improving prediction accuracy while maintaining a compact and interpretable feature subset.

Figure 8 illustrates the SHAP summary plot for the features selected by the BPSO algorithm, providing a global interpretation of their contributions to the CNN-DNN model's RUL predictions. The plot shows that SOH and BCt have the largest impact on the model output, as evidenced by their wide SHAP value distributions. Higher values of SOH and BCt (shown in red) contribute positively to RUL predictions, indicating longer remaining life, whereas lower values (shown in blue) are associated with negative SHAP values and reduced RUL estimates. Conversely, the cycle feature shows an inverse relationship: higher cycle counts tend to decrease the predicted RUL, reflecting cumulative degradation. Thermal and electrical features (chT, disT, and disV) make comparatively smaller but non-negligible contributions, capturing operational conditions that indirectly influence degradation behavior. Overall, the SHAP analysis confirms that the proposed model relies on physically meaningful features and provides transparent, interpretable insights into how individual variables affect RUL prediction.

BPSO, a discrete variant of the classical particle swarm optimization algorithm, is utilized in this study to perform wrapper-based feature selection for battery RUL prediction. As illustrated in Algorithm 1, each particle encodes a candidate feature subset using a binary position vector, while velocity updates govern the probabilistic inclusion or exclusion of features. The search process is driven by both individual best and global best solutions, enabling an effective balance between exploration and exploitation. A composite fitness function, combining the validation loss of the CNN-DNN predictor and a feature cardinality penalty, is employed to guide the optimization toward compact and informative feature subsets. Through iterative updates and swarm convergence, BPSO identifies an optimal subset of degradation-sensitive features, thereby enhancing prediction

accuracy, reducing computational complexity, and improving the generalization capability of the proposed RUL prediction framework.

CNN-DNN

CNN-DNN Regression Objective Function

Let

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N \quad (19)$$

denote the battery dataset, where $x_i \in \mathbb{R}^d$ is the selected feature vector and $y_i \in \mathbb{R}$ is the true RUL.

The CNN-DNN model is parameterized by weights and biases θ and produces a prediction:

$$\hat{y}_i = f_{\text{CNN-DNN}}(x_i; \theta) \quad (20)$$

The learning objective of the CNN-DNN regressor is to minimize the MSE over the training (or validation) set:

$$\mathcal{L}_{\text{CNN-DNN}}(\theta) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (21)$$

This loss function penalizes large prediction errors and is widely adopted for RUL regression due to its smoothness and sensitivity to degradation trends.

Feature subset representation in BPSO

In the BPSO framework, each particle represents a candidate feature subset using a binary vector:

$$\mathbf{b} = [b_1, b_2, \dots, b_d], b_j \in \{0, 1\}$$

where:

- $b_j = 1$ indicates that feature j is selected,
- $b_j = 0$ indicates that feature j is excluded.

The original feature vector x_i is reduced to:

$$x_i^{(b)} = x_i \odot \mathbf{b} \quad (22)$$

where $\odot$ denotes element-wise selection.

Wrapper-based fitness evaluation

For each particle (feature subset) $\mathbf{b}$, the CNN-DNN model is trained using only the selected features, and its predictive performance is evaluated on the validation set. To balance prediction accuracy and feature compactness, a composite fitness function is defined as:

$$J(\mathbf{b}) = \alpha \cdot \mathcal{L}_{\text{CNN-DNN}}(\mathbf{b}) + (1 - \alpha) \cdot \frac{\|\mathbf{b}\|_0}{d} \quad (23)$$

where:

- $\mathcal{L}_{\text{CNN-DNN}}(\mathbf{b})$ is the validation MSE obtained using the selected feature subset, $\|\mathbf{b}\|_0 = \sum_{j=1}^d b_j$ is the number of selected features, d is the total number of available features, $\alpha \in [0, 1]$ controls the trade-off between accuracy and feature reduction.

The optimization objective of BPSO is therefore:

$$\min_{\mathbf{b} \in \{0, 1\}^d} J(\mathbf{b})$$

Integration of CNN-DNN Within BPSO (Wrapper Mechanism)

During each BPSO iteration:

1. A particle generates a candidate feature subset $\mathbf{b}$.
2. The dataset is reduced to the selected features.
3. The CNN-DNN model is trained on the training set using $x_i^{(b)}$.
4. The validation MSE is computed and used in the fitness function.
5. The particle's personal best and the global best solutions are updated based on fitness comparison.

Through iterative updates of particle velocities and positions, BPSO converges toward a feature subset that minimizes prediction error while maintaining a compact and informative representation.

By embedding the CNN-DNN regression model directly within the BPSO optimization loop, the proposed framework performs true wrapper-based feature selection, ensuring that selected features are optimized specifically for the nonlinear learning behavior of the CNN-DNN architecture. This joint optimization

significantly improves prediction accuracy, reduces overfitting, and enhances generalization in lithium-ion battery RUL prediction.

The proposed methodology

This study proposes an integrated framework for accurate RUL prediction of lithium-ion batteries by combining BPSO for feature selection with a hybrid (CNN–DNN) regression model. The overall methodology consists of five main stages: data preprocessing, feature selection using BPSO, CNN-based feature extraction, DNN-based regression, and performance evaluation.

Figure 9 illustrates the overall workflow of the proposed CNN–DNN framework integrated with BPSO for lithium-ion battery RUL prediction. The methodology begins with data preprocessing, including cleaning, normalization, and dataset partitioning, followed by BPSO-based feature selection, where candidate feature subsets are iteratively evaluated using a CNN–DNN wrapper mechanism. In this stage, CNN layers automatically extract high-level representations from the selected features, while the DNN performs nonlinear regression to estimate RUL, and the validation error is fed back to guide the BPSO search process. Once convergence is achieved, the optimal feature subset is used to train the final CNN–DNN model, which is then applied to the test set for RUL prediction and performance evaluation using multiple regression metrics. This flowchart highlights the tight integration between feature optimization and deep learning, demonstrating how the proposed methodology balances accuracy, model complexity, and computational efficiency.

The integration between BPSO and the CNN–DNN model forms a wrapper optimization framework in which feature selection and model training are jointly optimized. During each BPSO iteration, the selected feature subset defined by a particle is used to train the CNN–DNN model, and the resulting validation error is fed back as the fitness score. This feedback mechanism allows BPSO to iteratively refine the feature subset by retaining informative degradation-related variables and discarding redundant or weakly informative features. As a result, the CNN–DNN model receives a reduced and more informative feature space, which improves learning efficiency, reduces overfitting risk, and enhances the overall predictive performance for battery Remaining Useful Life estimation.

Overall Framework

Let the preprocessed dataset be defined as:

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N \quad (24)$$

where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the input feature vector and $y_i \in \mathbb{R}$ is the corresponding RUL value. The objective of the proposed methodology is to learn a nonlinear mapping:

$$\hat{y}_i = f(\mathbf{x}_i) \quad (25)$$

that minimizes the prediction error between the estimated and actual RUL values.

BPSO-Based Feature Selection

To remove redundant and weakly informative features, BPSO is employed as a wrapper-based optimization technique. Each particle encodes a candidate feature subset as a binary vector:

$$\mathbf{b}_i = [b_{i1}, b_{i2}, \dots, b_{id}], b_{ij} \in \{0, 1\} \quad (26)$$

The velocity and position of each particle are updated according to:

$$v_{ij}(t+1) = \omega v_{ij}(t) + c_1 r_1 (p_{ij} - x_{ij}) + c_2 r_2 (g_j - x_{ij}) \quad (27)$$

$$S(v_{ij}) = \frac{1}{1 + e^{-v_{ij}}} \quad (28)$$

$$x_{ij}(t+1) = \begin{cases} 1, & \text{if } r < S(v_{ij}) \\ 0, & \text{otherwise} \end{cases} \quad (29)$$

where ω is the inertia weight, c_1 and c_2 are acceleration coefficients, and $r_1, r_2, r \in (0, 1)$ are random variables.

The fitness function guiding the BPSO search is defined as:

$$J(\mathbf{b}) = \alpha \cdot \text{MSE}(\mathbf{b}) + (1 - \alpha) \cdot \frac{\|\mathbf{b}\|_0}{d} \quad (30)$$

where $\|\mathbf{b}\|_0$ represents the number of selected features and α balances accuracy and feature reduction. The optimal feature subset is obtained from the global best particle.

CNN-Based Feature Extraction

After feature selection, the reduced feature vector is fed into the CNN module to automatically extract high-level representations. Let the selected feature matrix be $\mathbf{X}_s$.

The convolution operation in the CNN layer is defined as:

$$\mathbf{h}_k = \sigma(\mathbf{W}_k * \mathbf{X}_s + b_k) \quad (31)$$

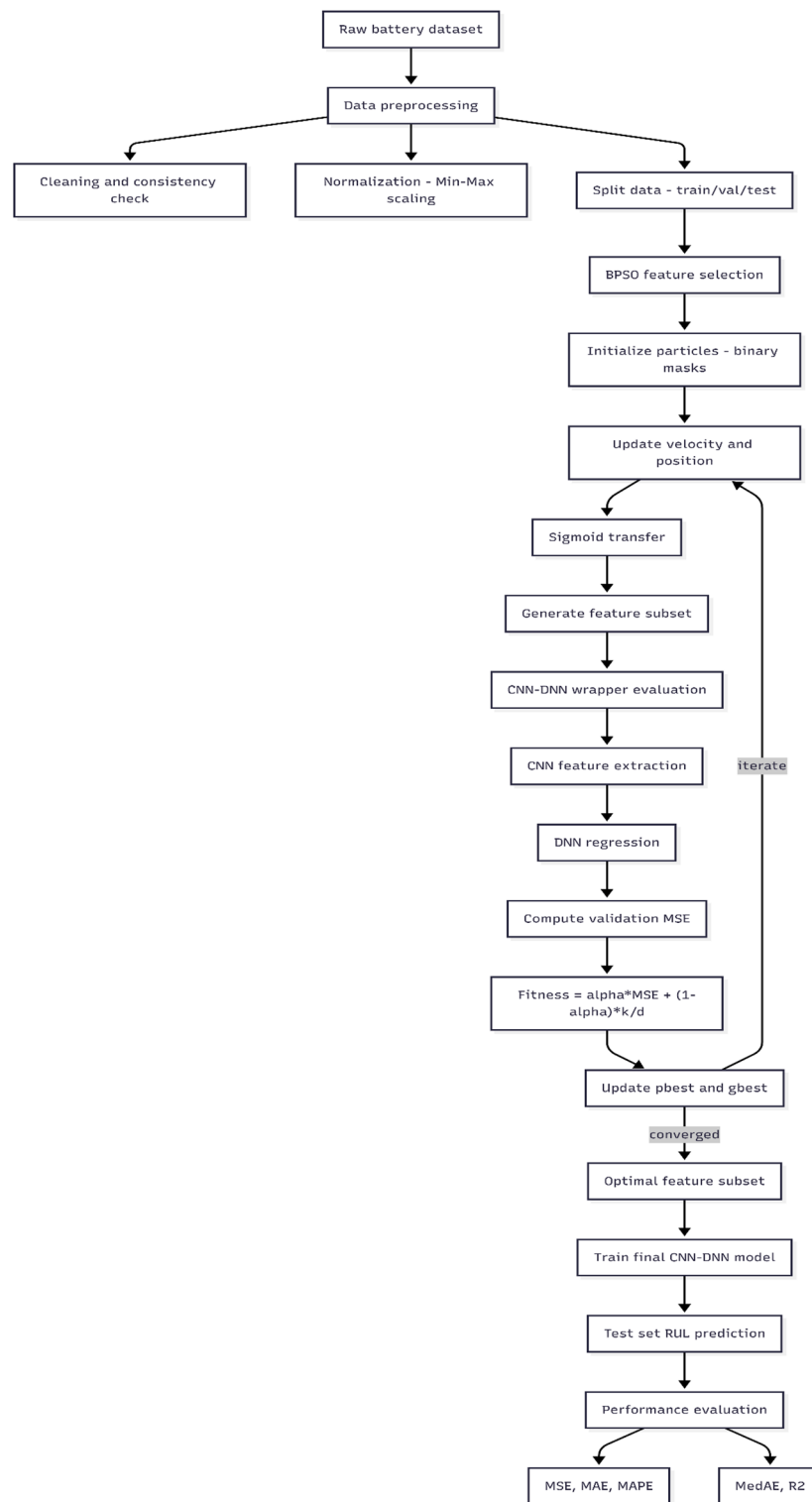


Fig. 9. Flowchart of the proposed CNN–DNN framework integrated with BPSO for lithium-ion battery RUL prediction.

where $*$ denotes convolution, $\mathbf{W}_k$ is the kernel matrix, b_k is the bias term, and $\sigma(\cdot)$ is the activation function (ReLU).

Pooling layers are applied to reduce dimensionality and improve robustness:

$$\mathbf{p}_k = \text{Pooling}(\mathbf{h}_k) \quad (32)$$

1. Input: Preprocessed dataset $\mathcal{D}$, BPSO parameters, CNN–DNN hyperparameters
2. Initialize BPSO particles and velocities
3. For each BPSO iteration:
 - Generate feature subset using binary position vectors
 - Train CNN–DNN using selected features
 - Compute validation MSE and fitness value
 - Update personal and global best particles
4. Obtain the optimal feature subset from the global best
5. Train the final CNN–DNN model using optimal features
6. Predict RUL on the test dataset
7. Output: Predicted RUL values and performance metrics

Algorithm 2: CNN–DNN with BPSO for Battery RUL Prediction.

The output of the CNN module is flattened and passed to the DNN regressor.

DNN-Based RUL Regression

The Deep Neural Network consists of multiple fully connected layers that model complex nonlinear relationships between extracted features and RUL. The forward propagation is defined as:

$$\mathbf{z}^{(l)} = \sigma(\mathbf{W}^{(l)}\mathbf{z}^{(l-1)} + \mathbf{b}^{(l)}) \quad (33)$$

The final output layer produces the predicted RUL:

$$\hat{y} = \mathbf{W}^{(L)}\mathbf{z}^{(L-1)} + b^{(L)} \quad (34)$$

The CNN–DNN model is trained by minimizing the Mean Squared Error loss:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (35)$$

Gradient-based optimization is used to update model parameters.

Algorithm 2 summarizes the integrated procedure of the proposed lithium-ion battery RUL prediction framework, which combines BPSO with a hybrid CNN–DNN regression model. The algorithm begins by initializing the BPSO population, where each particle encodes a candidate feature subset, and iteratively updates particle velocities and positions to explore the feature space. For each candidate subset, a CNN–DNN model is trained and evaluated using validation data, and the resulting prediction error is employed as a fitness measure to guide the optimization process. Through successive updates of personal and global best solutions, the algorithm converges toward an optimal set of degradation-sensitive features. Finally, the selected feature subset is used to train the final CNN–DNN model, which generates accurate RUL predictions on unseen test data. This integrated algorithm jointly optimizes feature selection and deep learning-based regression, thereby improving prediction accuracy, robustness, and generalization.

Experimental Setup Computing Environment

All experiments were implemented in a Python-based deep learning environment. The proposed CNN–DNN model and the wrapper-based BPSO feature selection were executed on the same platform to ensure fair comparison across all methods. Random seeds were fixed to improve reproducibility, and all reported results were computed using the held-out test set.

Table 4 summarizes the architecture configuration and training hyperparameters of the proposed CNN–DNN model used for lithium-ion battery RUL prediction. The convolutional layers are responsible for extracting informative feature patterns from the selected battery degradation variables, while the fully connected layers perform nonlinear regression to estimate the remaining useful life. The Adam optimizer with a learning rate of 0.001 was employed to ensure stable and efficient training. Dropout regularization was applied to the dense layers to reduce overfitting and improve generalization performance.

Table 5 summarizes the hyperparameter search ranges and the final selected configurations for the evaluated deep learning models. The grid search procedure evaluated multiple architectural and training configurations to ensure fair and consistent comparisons among the models. The selected hyperparameters correspond to the configurations achieving the lowest validation error, thereby providing balanced model complexity and prediction accuracy for lithium-ion battery RUL estimation.

Evaluation Metrics

Model performance was assessed using multiple regression metrics³⁵ to provide a comprehensive evaluation of error magnitude and goodness of fit:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (36)$$

Layer Type	Layer Description	Filters/Units	Kernel Size	Activation	Dropout	Additional Details
Input Layer	Battery feature vector input	10 features	–	–	–	Selected features from BPSO
Conv1D Layer 1	Feature extraction	32 filters	3	ReLU	–	Padding = same
MaxPooling1D	Down-sampling	–	2	–	–	Pool size = 2
Conv1D Layer 2	Deeper feature extraction	64 filters	3	ReLU	–	Padding = same
MaxPooling1D	Feature compression	–	2	–	–	Pool size = 2
Flatten Layer	Convert feature maps to vector	–	–	–	–	–
Dense Layer 1	Fully connected layer	128 units	–	ReLU	0.3	Regularization via dropout
Dense Layer 2	Fully connected layer	64 units	–	ReLU	0.3	–
Output Layer	Regression output	1 unit	–	Linear	–	Predicts Remaining Useful Life
Training Hyperparameters						
Parameter			Value			
Optimizer			Adam			
Learning Rate			0.001			
Batch Size			32			
Number of Epochs			100			
Loss Function			Mean Squared Error (MSE)			
Early Stopping			Enabled (patience = 10)			

Table 4. Architecture configuration and hyperparameters of the proposed CNN–DNN model for RUL prediction.

Model	Hyperparameter	Search Range	Selected Value
CNN	Number of convolution layers	{1, 2, 3}	2
	Filters per layer	{16, 32, 64}	32, 64
	Kernel size	{2, 3, 5}	3
	Learning rate	{0.01, 0.001, 0.0005}	0.001
DNN	Hidden layers	{2, 3, 4}	2
	Neurons per layer	{64, 128, 256}	128
LSTM	Number of LSTM layers	{1, 2}	1
	Units	{32, 64, 128}	64
GRU	Number of GRU layers	{1, 2}	1
	Units	{32, 64, 128}	64
BiGRU	Number of layers	{1, 2}	1
	Units	{32, 64, 128}	64
All models	Dropout rate	{0.2, 0.3, 0.5}	0.3
	Batch size	{16, 32, 64}	32
	Epochs	{50, 100, 150}	100
	Optimizer	{Adam, RMSprop}	Adam

Table 5. Hyperparameter search space and selected configurations for the evaluated deep learning models.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \tag{37}$$

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \tag{38}$$

$$MedAE = \text{median}(|y_i - \hat{y}_i|) \tag{39}$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \tag{40}$$

where $\bar{y}$ is the mean of ground-truth RUL values.

To provide a more rigorous evaluation of model generalization, battery-independent validation strategies are often employed in battery prognostics studies. One commonly used approach is leave-one-battery-out validation, where all cycles from one battery cell are excluded from the training data and used exclusively for testing. This ensures that the model is evaluated on completely unseen battery degradation trajectories.

Model	MSE	MAE	MAPE	MedAE ↓	R ² (%)
DNN	0.0990	0.1980	0.0302	0.1641	93.51
CNN	0.0542	0.1524	0.0236	0.1285	96.35
LSTM	0.1637	0.2618	0.0397	0.2234	89.78
GRU	0.2085	0.2984	0.0451	0.2597	87.34
BiGRU	0.2575	0.3329	0.0518	0.2986	84.84
Proposed CNN–DNN + BPSO	0.0141	0.0931	0.0142	0.0739	99.01

Table 6. Performance comparison of the proposed CNN–DNN model with BPSO-based feature selection against baseline deep learning models for lithium-ion battery RUL prediction.

Experiment	Feature Configuration	MSE	MAE	MAPE	MedAE	R ² (%)
CNN–DNN (without BPSO)	All available features	0.0218	0.1186	0.0197	0.0964	96.32
CNN–DNN + BPSO (Proposed)	Selected optimal feature subset	0.0141	0.0931	0.0142	0.0739	99.01

Table 7. Impact of BPSO-based feature selection on the CNN–DNN model performance for lithium-ion battery RUL prediction.

Although the present study focuses on evaluating the proposed framework under a balanced data partitioning strategy, future experiments will incorporate battery-wise evaluation protocols to further assess the robustness and transferability of the CNN–DNN model across different battery cells and operating conditions.

Results and Discussion

The performance of the proposed CNN–DNN model integrated with BPSO was evaluated on the test dataset using multiple regression metrics, including MSE, MAE, MAPE, MedAE, and R^2 . The results demonstrate that the proposed framework achieves high prediction accuracy and strong generalization capability, indicating its effectiveness in modeling the nonlinear degradation behavior of lithium-ion batteries. The low error values and high coefficient of determination confirm that the predicted RUL closely matches the ground-truth values across different degradation stages.

Table 6 presents a quantitative comparison of the proposed CNN–DNN model integrated with BPSO against several baseline deep learning models for lithium-ion battery RUL prediction. As shown in the table, the proposed method significantly outperforms all comparative models across every evaluation metric, achieving the lowest MSE (0.0141), MAE (0.0931), MAPE (0.0142), and MedAE (0.0739), while attaining the highest coefficient of determination ($R^2 = 99.01\%$). In contrast, standalone CNN and DNN models exhibit higher prediction errors and lower goodness-of-fit, whereas recurrent architectures such as LSTM, GRU, and BiGRU show further performance degradation. These results clearly demonstrate that the joint integration of CNN-based feature extraction, DNN-based nonlinear regression, and BPSO-driven feature selection substantially enhances prediction accuracy, robustness, and generalization capability for battery RUL estimation.

To further investigate the impact of the BPSO-based feature selection process, a feature sensitivity analysis was conducted to evaluate how different battery degradation variables influence the model predictions. The selected features were analyzed using SHAP, which provides a unified framework for interpreting machine learning models by quantifying the contribution of each feature to the prediction outcome. The SHAP analysis reveals that certain degradation indicators, such as voltage variation, discharge capacity, and internal resistance, have a stronger influence on the predicted Remaining Useful Life compared with other variables. This observation is consistent with the physical degradation behavior of lithium-ion batteries, where these parameters reflect the aging state and capacity fading of the battery.

Table 7 presents the impact of the BPSO-based feature selection process on the predictive performance of the CNN–DNN model. The results indicate that applying feature selection significantly improves the model performance across all evaluation metrics. Specifically, the MAE decreases from 0.1186 to 0.0931 and the MSE decreases from 0.0218 to 0.0141 after applying BPSO. In addition, the (R^2) increases from 96.32% to 99.01%, demonstrating that the selected feature subset allows the model to capture battery degradation patterns more effectively. These findings confirm that integrating BPSO with the CNN–DNN architecture enhances prediction accuracy while reducing redundant input information.

Table 8 presents the comparison between the proposed CNN–DNN model and classical regression methods. The results indicate that Linear Regression exhibits the lowest prediction accuracy due to its limited capability in modeling nonlinear degradation patterns. Random Forest improves performance by leveraging ensemble learning; however, it still performs worse than the proposed hybrid deep learning framework. The CNN–DNN model combined with BPSO-based feature selection achieves the best performance across all evaluation metrics, highlighting the effectiveness of deep feature extraction and optimized feature selection in improving RUL prediction accuracy.

Figure 10 illustrates the convergence behavior of the MAE during training and testing for the proposed CNN–DNN model and the comparative deep learning baselines, including CNN, DNN, LSTM, GRU, and BiGRU. Across all models, MAE decreases rapidly during the initial training epochs, indicating effective learning

Model	MSE	MAE	MAPE	MedAE	R^2 (%)
Linear Regression	0.0456	0.2417	0.0328	0.1815	92.14
Random Forest	0.0283	0.1679	0.0213	0.1184	95.87
Proposed CNN–DNN + BPSO	0.0141	0.0931	0.0142	0.0739	99.01

Table 8. Comparison of the proposed model with classical regression methods.

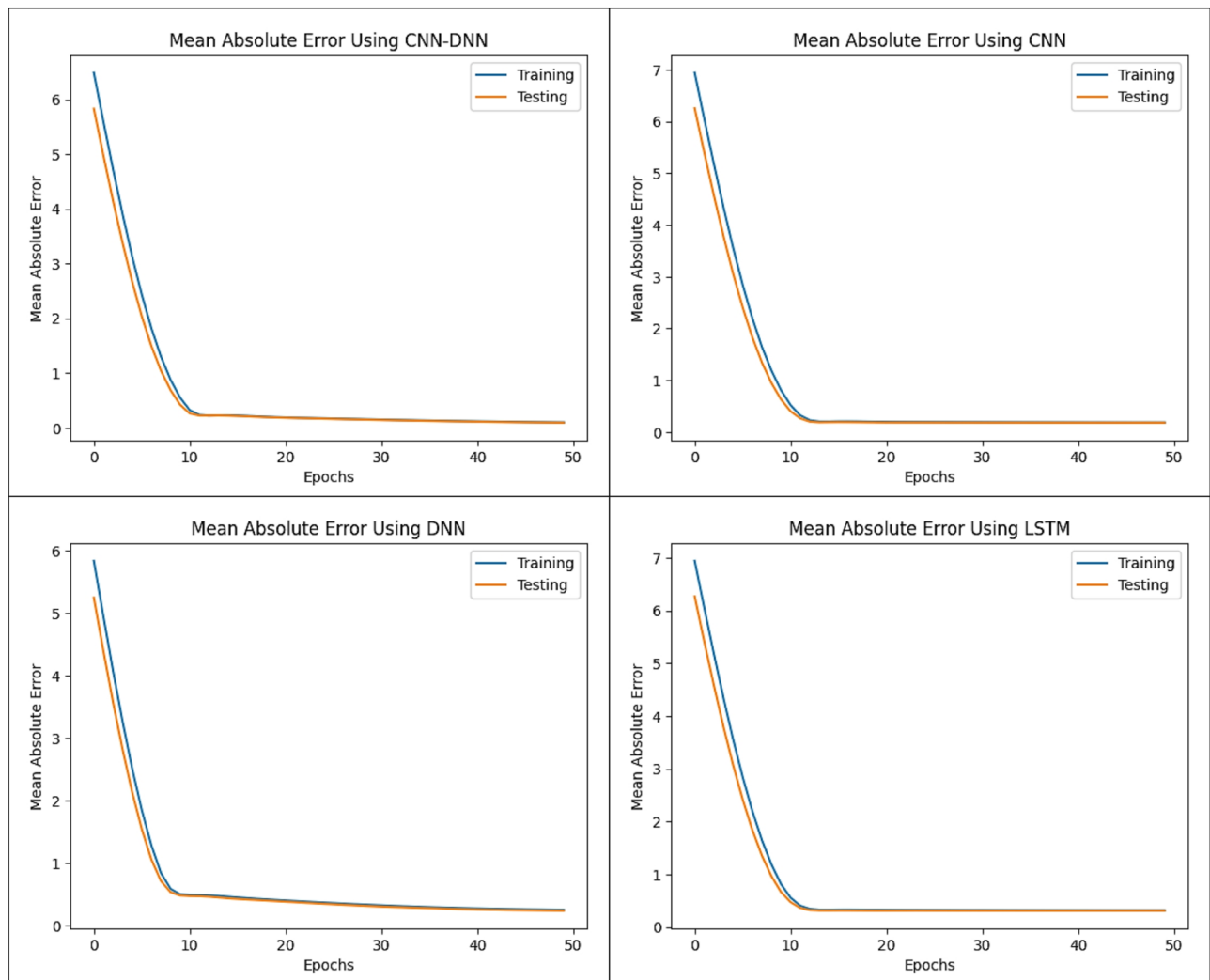


Fig. 10. Training and testing MAE convergence curves for the proposed CNN–DNN model and baseline deep learning models over successive training epochs.

of degradation patterns from the battery data. The proposed CNN–DNN model exhibits the fastest convergence and the lowest final MAE values on both the training and test sets, demonstrating superior learning efficiency and strong generalization. Moreover, the close alignment between training and testing curves for the CNN–DNN model suggests minimal overfitting, whereas recurrent models show comparatively higher residual errors before stabilization. These results confirm that integrating convolutional feature extraction with deep nonlinear regression enhances both convergence stability and predictive accuracy for lithium-ion battery RUL estimation.

Figure 11 depicts the training and testing MSE convergence curves for the proposed CNN–DNN model and the comparative deep learning approaches, including CNN, DNN, LSTM, GRU, and BiGRU. All models demonstrate a rapid reduction in MSE during the early training epochs, indicating effective learning of battery degradation patterns. However, the CNN–DNN model converges more quickly and stabilizes at a significantly lower MSE on both the training and test datasets, reflecting superior predictive accuracy and robustness. The close overlap between training and testing curves further indicates strong generalization and minimal overfitting. In contrast, recurrent models exhibit higher residual errors before convergence, suggesting that they are less effective at capturing heterogeneous feature interactions than the hybrid CNN–DNN architecture.

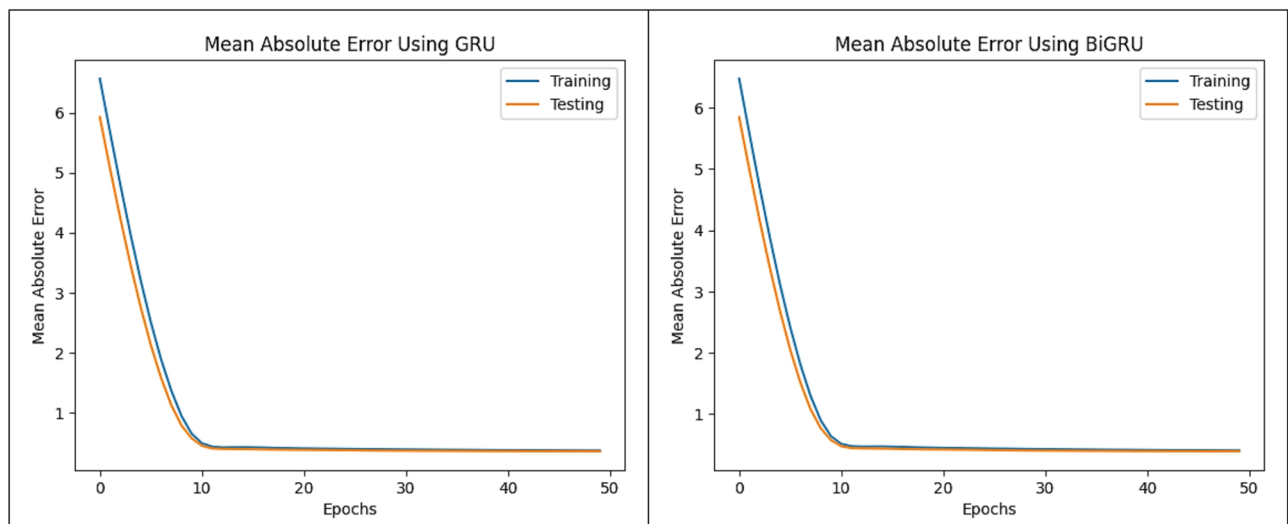


Fig. 10. (continued)

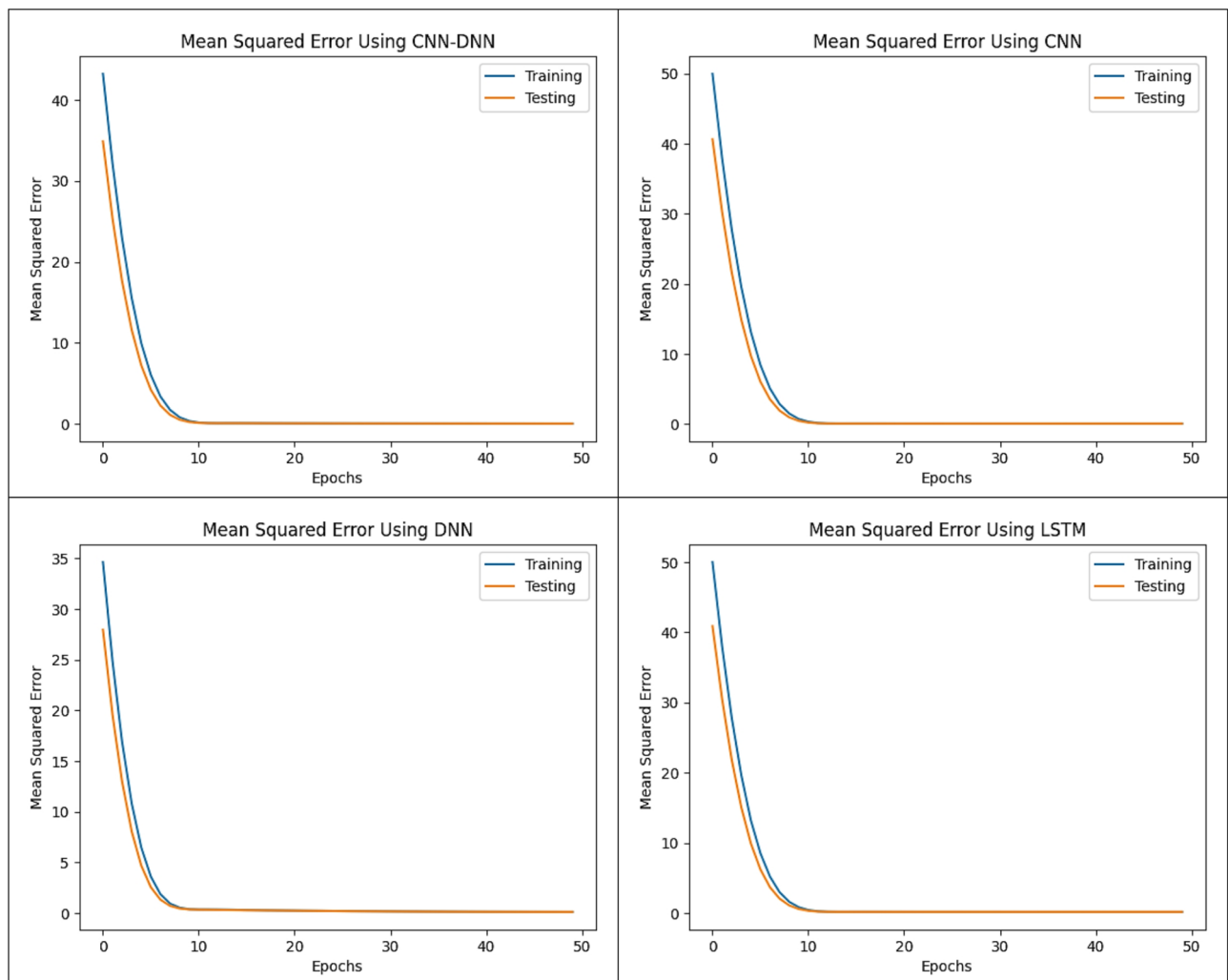


Fig. 11. Training and testing MSE convergence curves for the proposed CNN–DNN model and baseline deep learning models across training epochs.

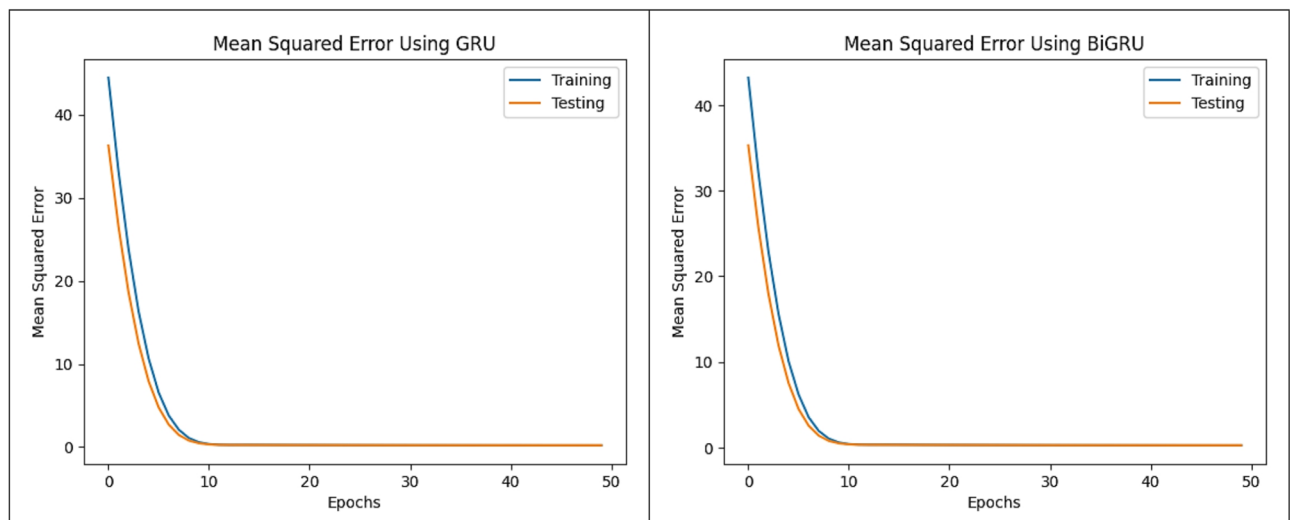


Fig. 11. (continued)

These observations confirm the effectiveness of the proposed model in minimizing squared prediction errors for lithium-ion battery RUL estimation.

Figure 12 presents the scatter plots of predicted versus actual Remaining Useful Life (RUL) for the proposed CNN–DNN model and the comparative baseline models, including CNN, DNN, LSTM, GRU, and BiGRU. Each subplot illustrates the degree of agreement between predicted and true RUL values, along with a fitted regression line and the corresponding coefficient of determination (R^2). The proposed CNN–DNN model demonstrates the strongest linear agreement, achieving an R^2 of approximately 99.01%, with prediction points tightly clustered around the ideal diagonal line, indicating highly accurate and consistent predictions. In contrast, the standalone CNN and DNN models show moderate dispersion with lower R^2 values, while recurrent architectures exhibit noticeably larger deviations from the fitted line, reflecting reduced prediction accuracy. These visual comparisons clearly confirm the superiority of the proposed CNN–DNN framework in capturing the nonlinear degradation behavior of lithium-ion batteries and delivering precise RUL predictions.

Statistical analysis

To further verify the robustness of the obtained results, a statistical significance analysis was conducted between the proposed CNN–DNN model with BPSO and the baseline deep learning models using a paired t-test. The comparison was performed using the absolute prediction errors derived from the test dataset, and the significance level was set to $\alpha=0.05$. Table 9 summarizes the statistical comparison results. The positive mean error differences indicate that the proposed model consistently achieved lower prediction errors compared with the baseline models.

Comparative analysis

Table 10 compares the proposed CNN–DNN model optimized using BPSO with a recent state-of-the-art study that employs the same lithium-ion battery degradation dataset. Vilsen and Stroe³⁶ demonstrate that deep neural networks outperform classical regression models when features are extracted from partial charging intervals; however, it primarily focuses on state-of-health estimation rather than remaining useful life prediction. In contrast, the proposed method directly targets RUL forecasting and integrates automatic feature selection with a hybrid deep architecture, resulting in significantly lower prediction errors and a higher coefficient of determination. This comparison highlights the effectiveness of combining CNN-based feature extraction with DNN regression and metaheuristic optimization for accurate RUL prediction under realistic battery degradation conditions.

Table 11 presents a comparison between the proposed CNN–DNN model with BPSO-based feature selection and several representative recent approaches for lithium-ion battery RUL prediction. The comparison indicates that most recent studies employ hybrid architectures that combine convolutional feature extraction with sequential or optimization-based models. The proposed approach follows a similar hybrid strategy but introduces a wrapper-based feature selection mechanism using BPSO, which improves the learning efficiency of the CNN–DNN architecture. The results demonstrate that the proposed framework achieves competitive performance while maintaining a relatively lightweight and computationally efficient model design.

Conclusion and future work

This study presented a hybrid CNN–DNN framework integrated with BPSO for accurate RUL prediction of lithium-ion batteries using a realistic degradation dataset. The results demonstrate that combining convolutional feature extraction with deep regression, guided by wrapper-based feature selection, enables the model to effectively capture both local nonlinear patterns and long-term degradation trends. Compared with standalone

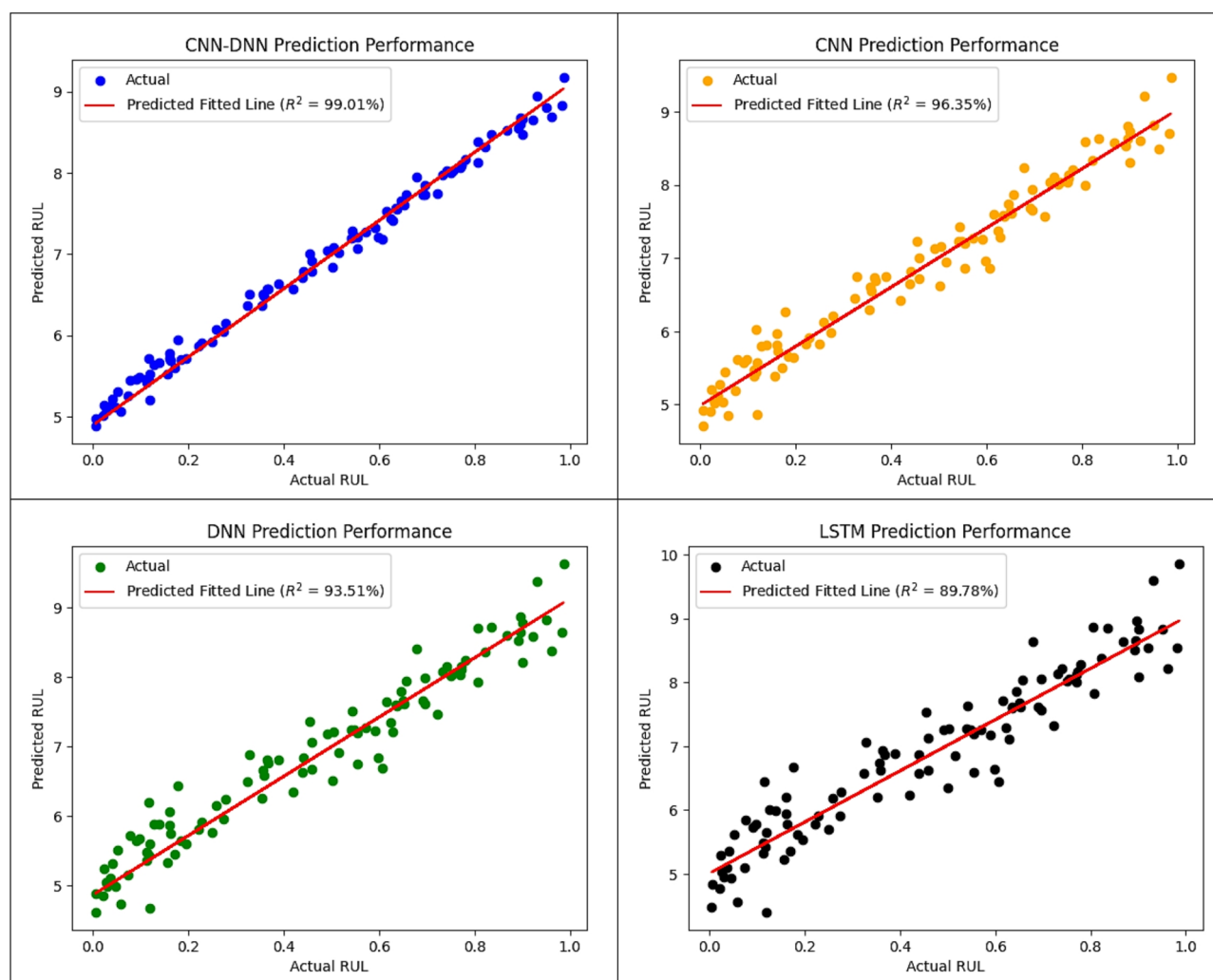


Fig. 12. Predicted versus actual RUL comparison for the proposed CNN–DNN model and baseline deep learning models, including fitted regression lines and R^2 values.

deep learning and recurrent models, the proposed approach consistently achieved lower error metrics and a higher coefficient of determination, confirming the benefit of jointly optimizing feature relevance and model learning capacity. Beyond predictive accuracy, the analysis of feature importance and SHAP explanations revealed the dominant influence of health-related and cycling features on battery aging behavior, thereby enhancing the interpretability and practical relevance of the model for battery management systems.

While the proposed CNN–DNN model has shown good predictive results on the used lithium-ion battery degradation dataset, it is also important to note that the used dataset has only 680 samples, which are cycle-level measurements from only a few battery cells. While such dataset sizes are common for battery prognostics research studies because of the long duration needed for battery degradation tests, the proposed model's results and performance can be further improved and validated on other publicly available datasets for battery prognostics. In future works, the proposed model will be tested on larger battery prognostics datasets, such as the NASA Prognostics Center of Excellence battery dataset, CALCE battery dataset, and the Stanford-MIT battery life dataset, which include battery measurements from different battery chemistries, charge-discharge modes, and environmental conditions.

Although the proposed CNN–DNN framework integrated with BPSO-based feature selection demonstrates promising performance for lithium-ion battery Remaining Useful Life prediction, several important research directions remain for future investigation. One key direction is the optimization of the proposed model for real-time battery health monitoring applications. In practical battery management systems (BMS), predictive models must operate efficiently under limited computational resources while processing streaming sensor data. Therefore, future work will explore lightweight deep learning architectures and model compression strategies that enable deployment in edge devices and embedded systems.

Recent advances in efficient deep learning architectures have shown that combining feature extraction networks with optimized model structures can significantly improve computational efficiency while maintaining high prediction accuracy. For example, lightweight hybrid deep learning models have been successfully applied

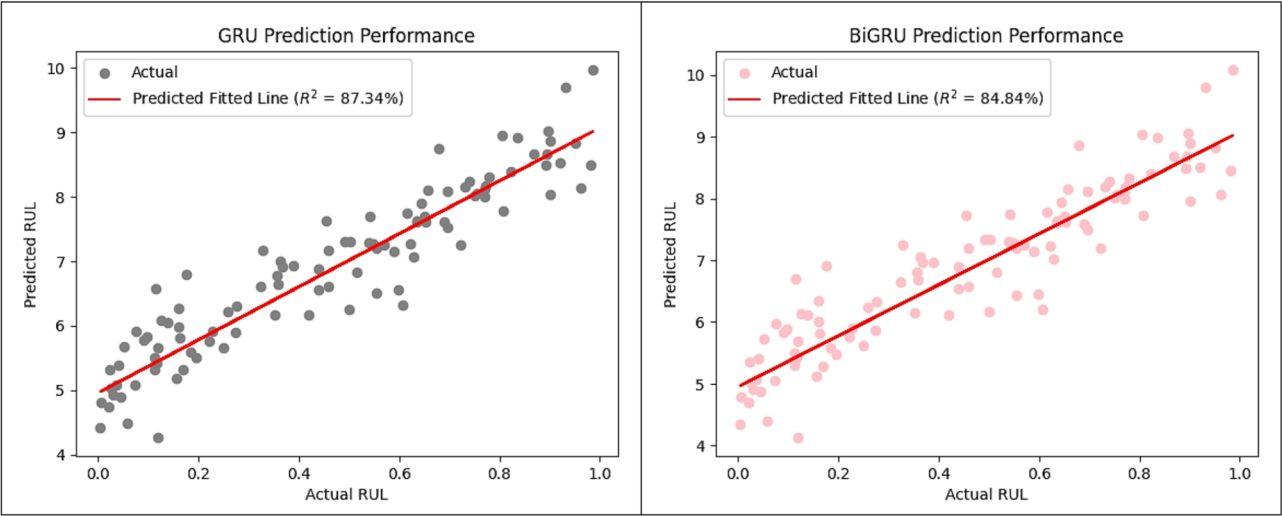


Fig. 12. (continued)

Compared Models	Mean Error Difference (MAE_baseline – MAE_proposed)	t-statistic	p-value	Significance (α=0.05)
Proposed vs. DNN	0.1049	4.21	0.0015	Significant
Proposed vs. CNN	0.0593	3.48	0.0038	
Proposed vs. LSTM	0.1687	5.03	0.0006	
Proposed vs. GRU	0.2053	5.67	0.0002	
Proposed vs. BiGRU	0.2398	6.12	0.0001	

Table 9. Statistical significance analysis between the proposed CNN–DNN + BPSO model and baseline models using paired t-test on MAE errors.

Study	Prediction Task	Model	Feature Strategy	Reported Performance
Vilsen and Stroe [36]	SOH estimation	MLR, SVR, RF, Deep NN	Partial charging interval features; cycle index (FEC) analysis	NN outperforms classical models; performance differences ≤ 0.41% across feature scenarios
This work	RUL prediction	CNN–DNN + BPSO	Wrapper-based optimal feature selection (SOH, BCT, cycle, thermal & electrical features)	MSE = 0.0141, MAE = 0.0931, MAPE = 0.0142, R² = 99.01%

Table 10. Comparative analysis between the proposed CNN–DNN with BPSO-based feature selection and a recent deep learning study using the same lithium-ion battery degradation dataset.

Study	Year	Method	Idea	Performance
Chen et al. [37]	2022	CNN–LSTM	CNN feature extraction + temporal modeling	Improved prediction accuracy over single models
Mou et al. [38]	2024	1D-CNN + BLSTM	CNN deep features with sequential modeling	Improved stability and accuracy
Guo et al. [39]	2023	CEEMDAN–CNN–BiLSTM	Signal decomposition + hybrid deep learning	Higher accuracy than CNN or BiLSTM alone
Fan et al. [40]	2025	SVR–LSTM + optimization	Hybrid ML + optimization algorithm	R² > 0.96 on NASA dataset
Gao et al. [41]	2025	parallel Transformer–BiGRU–GlobalAttention model	The parallel Transformer encoder captures long-range temporal dependencies, while the BiGRU network enhances local sequence associations through bidirectional modeling. The global attention mechanism dynamically weights key temporal features, and cross-attention facilitates spatiotemporal feature interaction and fusion.	MSE of 0.078, MAE of 0.233, and RMSE of 11.13
Proposed Model	This Study	CNN–DNN + BPSO	Feature selection + hybrid deep learning	MAE = 0.0931, MSE = 0.0141, R² = 99.01%

Table 11. Comparison of the proposed model with representative recent RUL prediction approaches.

in complex recognition and prediction tasks through the integration of optimized convolutional networks and neural architecture search techniques^{42,43}. Similarly, hybrid deep learning frameworks have demonstrated promising performance in battery Remaining Useful Life prediction by combining different neural network structures to capture degradation patterns more effectively⁴⁴. Inspired by these developments, future studies may explore compact architectures, neural architecture search, and edge-optimized deep learning models to improve real-time deployment capabilities⁴⁵.

Another important research direction is the exploration of alternative feature selection and dimensionality reduction techniques to further improve model efficiency and robustness. While BPSO demonstrated strong performance in identifying informative features, other optimization algorithms such as genetic algorithms, ant colony optimization, or hybrid metaheuristic methods may provide additional improvements in feature subset stability and prediction performance. Investigating these techniques may lead to more computationally efficient models while maintaining high prediction accuracy.

Data availability

The data that support the findings of this study are available at <https://www.kaggle.com/datasets/programmer3/lithium-ion-battery-degradation-dataset/data>.

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Author contributions

Hala khoufi: Writing, Project administration, and Conceptualization. Emna Bouazizi: Writing, Investigation, and Project Administration. Ahmed M. Elshewey: prepared figures and recorded the table results. Ayman E. Khedr: Wrote the main manuscript, edited, and performed Formal analysis. All authors reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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